Controlled Relaxation Protocols for the Wamura Problem

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Abstract

A restriction justified by a long-timeline risk cannot be revised through observation: the evidence that would revise it cannot arrive within any practical decision cycle, and waiting for the catastrophe that would generate the evidence is the failure mode the restriction was meant to prevent. We call this the *Wamura problem*. We formalize the structural asymmetry, prove that max-aggregation with neutral-prior initialization produces monotone overcaution absent external intervention, and introduce *controlled relaxation protocols*: bounded, monitored exercise windows that generate genuine Bayesian evidence about risk claims while provably bounding maximum damage. A scope-construction theorem bounds the damage from a test by a user-set parameter ϵ even when cross-boundary cooperative capabilities could propagate the test's effects. A convergence theorem shows that repeated controlled relaxation drives risk estimates almost surely to truth at exponential rate, with the rate given by the Kullback–Leibler divergence between the observation distributions. We extend the protocol to *keystone-refinement tests*, which handle the common case where a restriction is not obsolete but misconfigured: the underlying risk pathway still operates, but technological progress has made narrower keystones available that would block it without blocking adjacent benign pathways. For the subset of risks admitting bounded-scope feasibility (a precondition we delineate in Section 11), the protocol closes the most important open problem named in [Lasser, 2026c] and resolves the degenerate-phase-boundary case of [Lasser, 2026d, Proposition 4].

1 Introduction

Escalating capabilities within a society create escalating mechanisms for disaster the civilisation has never experienced, some of which carry existential risk if pursued without restraint. This capability-escalating regime is precisely the $\mathrm{vol}_{\mathrm{P}}$-maximising dynamic that Goal-Frontier Maximization (GFM) seeks to produce; the question this paper addresses is how to balance the existential risks of such escalation against the reciprocal danger of a society that becomes increasingly bound by restrictive overcaution, accumulating restrictions it cannot revise through observation, approaching the over-calibrated-restriction feedback loop of [Lasser, 2026d, Loop 3].

The classical balancing mechanism is catastrophe-as-teacher: a first occurrence writes historical markers—tsunami stones, disaster records, institutional memory, building-code revisions, pharmacovigilance regimes—that subsequent generations consult in lieu of the experiment they cannot run. Disaster-policy research has long studied this dynamic under the heading of focusing events [Birkland, 1997]: the policy attention required to revise restrictions accumulates only after the catastrophe that made the original restriction look prescient. In the AI-safety literature the same structural concern appears as *safe exploration*—an agent cannot learn the consequences of an action it never takes [Amodei et al., 2016]—and as the gradual-misalignment scenario of Christiano [2019], in which incremental restrictions accumulate without counter-evidence and the resulting failure looks less like a catastrophic single event than like the slow erosion of a society's effective optionality. The

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precautionary-principle literature in policy and law makes the same structural tradeoff explicit: precaution against unfamiliar harms is appropriate but accumulates costs the precaution itself renders invisible [Sunstein, 2005]. This paper formalises the structural problem common to these literatures and constructs a mechanism for generating the evidential content historical markers would have recorded, within a bounded damage budget chosen by the framework rather than the next generation.

The specific failure mode in the existing GFM framework is the max-aggregation pathology of [Lasser, 2026c], named there as the most important open problem of the sequence: the conservative max operator, combined with the verification asymmetry inherent in risk prevention, produces overcaution that is structurally irresolvable within the existing machinery—a risk claim that successfully prevents catastrophe generates no observation that could lower the risk estimate, because the very success of the restriction is what prevents the counterfactual test. This is a distinct problem from the compound feedback loops analysed in [Lasser, 2026d]: that work asks when partial subsumption self-corrects versus cascades, while the present paper asks a complementary question—when the framework identifies a risky capability and imposes a restriction, how does it learn whether the restriction remains necessary, or whether technological progress has mitigated the underlying risk? The two problems reinforce each other: [Lasser, 2026d, Proposition 4] shows the restriction-verification gap is a degenerate phase boundary with zero self-correction rate ($r_{\mathrm{S}} = 0$), but the phase-boundary machinery does not by itself supply the missing self-correction mechanism.

The paper is named for Mayor Kotoku Wamura's tsunami-preparedness legacy at Fudai [Lasser, 2026e, Discussion]: an oversized seawall built on the basis of centuries-old stone markers warning against construction below a certain elevation, ridiculed for decades as wasteful overcaution, and vindicated when the 2011 Tohoku tsunami reached precisely the elevation the markers had recorded. The Fudai case describes the shape of the problem; the question this paper answers is how to have Wamura's foresight without the ancient guidance that informed him.

1.1 The structural problem: long-timeline risks

The ratchet arises from a timeline mismatch between the risk and the framework's observational window. Let T be the realisation timeline of a risk—the interval on which the risk would actually manifest under exercise—and W be the decision-cycle observational window over which the framework can accumulate evidence. When $T \ll W$, observation resolves the question promptly; the restriction dynamics are well-behaved. When $T \gg W$, no observation can update the risk estimate within any practical decision cycle. The restriction is adopted under uncertainty, survives because no counter-evidence can arrive in time to challenge it, and restrictions of this class accumulate monotonically because: (i) new long-timeline risks are continually discovered, each imposing a restriction; (ii) existing long-timeline restrictions cannot be revised through observation. The ratio of restriction-imposing decisions to restriction-revising decisions approaches infinity on the long-timeline subset.

Three mechanisms compose with the timeline mismatch to produce the pathology.

Max-aggregation. The risk framework of [Lasser, 2026c] takes the maximum over agents' risk estimates, so a single high-risk claim from any agent constrains the capability tuple. This is conservative by design—it prevents the framework from dismissing minority risk warnings—but it means a single overly cautious agent can lock a capability indefinitely when the underlying risk is long-timeline.

Verification asymmetry. The monotonicity result [Lasser, 2026a, Proposition 1] shows that as an agent's capabilities grow, the verification asymmetry between what it can demonstrate and what can be externally verified is non-decreasing. Applied to risk claims: verifying a risk (observing catastrophe) is operationally straightforward on any timeline; verifying safety (observing non-catastrophe under exercise) requires the exercise to happen, which the restriction prevents, and even when permitted would not produce a conclusion within W when $T \gg W$.

Neutral-prior bootstrap. When a new capability is discovered and no exercise history exists, the framework initializes with a neutral prior [Lasser, 2026c, Discussion]. Combined with max-

aggregation, a single agent assigning high risk to the new capability immediately restricts it, and the restriction then prevents the evidence accumulation that would revise the estimate.

The three mechanisms compose into a ratchet: risk claims can raise p_R but the restricted regime provides no mechanism to lower it within the framework's operating horizon. We formalize this ratchet in Section 2, introduce controlled relaxation protocols as the structural resolution in Section 3, and prove the damage-bound and convergence theorems in Sections 4–6.

1.2 Framing precondition: $\mathrm{vol}_{\mathrm{P}}$ as an operational target

All results in this paper are stated in terms of $\mathrm{vol}_{\mathrm{P}}$ dynamics and Bayesian updates on risk claims about $\mathrm{vol}_{\mathrm{P}}$-contraction. We assume throughout that $\mathrm{vol}_{\mathrm{P}}$ is an operationally meaningful target, in the sense inherited from [Lasser, 2026b,d]: improving $\mathrm{vol}_{\mathrm{P}}$ corresponds to an improvement in the experiential capability volume the framework is designed to protect. This is the precondition under which the damage-bound and convergence claims carry their safety interpretation. If $\mathrm{vol}_{\mathrm{P}}$ diverges from the experiential target (the B-to-C gap [Lasser, 2026b]), the damage tolerance ϵ bounds contraction on the wrong quantity and every subsequent result inherits the divergence. Controlled relaxation does not close the B-to-C gap; it resolves the restriction-to-risk feedback gap that exists *above* the B-to-C precondition.

1.3 Summary of results

1. **Formalization of the Wamura asymmetry** (Section 2). We define risk-restriction pairs and their two observation regimes, then prove that under the restricted regime the Bayesian update has trivial likelihood ratio: successful prevention is observationally indistinguishable from unnecessary prevention (Proposition 1). Combined with max-aggregation and neutral-prior initialization, this produces monotone overcaution (Proposition 2) and an asymptotic lockup corollary.
2. **Controlled relaxation protocol** (Section 3). A controlled relaxation test lifts the restriction within a bounded sub-poset S for a fixed duration τ_{test}, under continuous multi-channel monitoring, with an instantaneous-damage safety cutoff at the tolerance ϵ. The test is formalized as a simple binary hypothesis problem (H_0: pathway harmless, H_1: pathway dangerous with fixed activation probability θ_1) with fixed per-test likelihood ratios $L_0 < 1 < L_1$.
3. **Damage bound** (Section 4). Theorem 1 bounds the worst-case $\mathrm{vol}_{\mathrm{P}}$ contraction during a test by ϵ during the test window and 2ϵ post-reinstatement, under a scope construction (Definition 4) whose cooperative-closure condition explicitly accounts for cross-boundary forward-reachability.
4. **Convergence theorem** (Section 6). Theorem 2 shows that under repeated controlled relaxation with fixed parameters, the posterior p_R converges almost surely to the correct hypothesis, with exponential rate given by the Kullback–Leibler divergence between the observation distributions under H_0 and H_1. Finite-sample corollaries give explicit bounds on the number of tests required to clear a false risk claim.
5. **Integration** (Section 10). The protocol preserves the self-balancing property of $\mathrm{vol}_{\mathrm{P}}$ (Proposition 7), provides empirical arbitration for inter-agent risk disagreement (Proposition 8), and extends the multi-channel detection architecture of [Lasser, 2026f] with a sixth, actively generated channel (Lemma 2).

1.4 Real-world interpretation

The formal construction of this paper is a programmatic mechanism for empirical review of long-timeline restrictions that may have outlived their justifying risk model. The framework identifies restrictions that persist without recent confirming evidence, formulates the underlying risk pathway that the restriction was intended to block, and designs a bounded-damage empirical test that probes whether the physical mechanism underlying the restriction still operates—a test conducted by appropriately trusted actors within a bounded scope, with the trust apparatus (cryptographic [Lasser, 2026a], governmental, institutional, legal, or otherwise) selected to match the domain.

Application domains where long-timeline restrictions accumulate and benefit from programmatic review:

- **Environmental contamination controls**: chemical scheduling where bioaccumulation or persistence timelines are multi-decade (DDT, PFAS, ozone-depleting refrigerants); the current restriction may target a mechanism that has changed under reformulation or that modern mitigation has obsoleted.
- **Nuclear dual-use controls**: export and technology controls where weapons-grade and reactor-grade pathways share keystone capabilities in broad restriction regimes designed under mid-20th-century technology. Modern enrichment and machining have differentiated to a degree that supports narrower keystone choices; see Section 9.
- **Pandemic preparedness and dual-use biology**: research moratoria on gain-of-function work, where the risk timeline is the interval between pandemic events (decades) and the observational window is research-cycle short.
- **Infrastructure safety margins**: dam, bridge, and seismic design codes originally calibrated against century-scale events, where construction material and mitigation-technology progress may permit refined margins.
- **AI capability restrictions**: the sequence's primary application. Restrictions adopted against concerns about future outcomes when no empirical counter-evidence can arrive within training and deployment cycles.

This list is illustrative; the formal content does not commit to any particular domain. The framework is indifferent to whether the trusted actors are human oversight bodies, cryptographically committed automated systems, or combinations thereof; the implementation of trust is a domain-specific design decision outside the scope of the formal construction.

A practical asymmetry across these domains is worth naming explicitly. Three of the four (chemical contamination, nuclear dual-use, pandemic preparedness) sit on top of pre-existing institutional infrastructure built for exactly this kind of bounded empirical test: the EPA, the IAEA, biosafety committees and BSL-tiered containment. The fourth, AI capability restriction, has no equivalent infrastructure for the bounded empirical testing this protocol assumes. The protocol's deployability in the AI case is therefore conditional on institutional development that this paper does not provide and does not estimate the cost of. We claim the formal apparatus, not the institutional readiness; the gap between the two is load-bearing for the AI application and should be acknowledged rather than elided.

1.5 What this paper does not do

- **It does not replace max-aggregation.** Log-odds risk aggregation [Lasser, 2026f, Definition 5] already provides an alternative for the multi-agent case. The verification asymmetry we resolve persists under *any* aggregation rule whenever a risk claim prevents the exercise that would generate evidence. Controlled relaxation is orthogonal to the choice of aggregator.
- **It does not address observation-channel integrity under adversarial conditions.** The commitment protocols of [Lasser, 2026a] handle that. We assume honest observation channels and honest (but possibly miscalibrated) risk assessors throughout.
- **It does not resolve the B-to-C gap.** Controlled relaxation generates evidence about $\mathrm{vol}_{\mathrm{P}}$-contraction claims, not about experiential value. Convergence of p_R to truth translates into convergence of the restriction landscape only insofar as $\mathrm{vol}_{\mathrm{P}}$ tracks the experiential target; the precondition from Section 1.2 is inherited.
- **It does not authorise unconstrained exercise of dangerous capabilities.** A controlled relaxation test occurs strictly within a bounded scope S, under continuous detector monitoring, with an instantaneous-damage cutoff at the user-set tolerance ϵ—a clinical-trial protocol, a contained laboratory, a sandboxed deployment, an instrumented test range, or other domain-appropriate constrained environment. The framework does not propose detonating bombs occasionally to confirm bombs still explode: risks with abundant current evidence are screened out by the meta-policy of Section 7 before any test is initiated, because the information value of confirming an established mechanism is near zero while the

expected damage remains positive. The protocol addresses the complementary case, where evidence is absent and restrictions accumulate without challenge.

- **It does not claim that controlled relaxation is always preferable to structural resolution.** Section 7 places the test-initiation decision against the information value of a structural argument [Lasser, 2026c, Proposition 2]: when a structural resolution is available (for example, a proof that the pathway is mechanically infeasible), it is strictly preferable to any controlled relaxation test.
- **It does not address the Fudai case directly.** At Fudai, historical evidence (ancient tsunami stones) existed but was dismissed in favour of the modern observational window. The remedy there is to restore historical records to the decision process, not to generate new experimental evidence—controlled relaxation would have been actively dangerous at Fudai (a test of a reduced seawall works perfectly until the tsunami arrives). The paper's domain is the complementary case where no historical record is available.

A table summarising the prior-paper results this paper relies on is collected in Appendix C.

2 The Verification Asymmetry

This section isolates the mechanism that makes the max-aggregation pathology structurally irresolvable within the existing risk framework, by formalizing the two observation regimes induced by a risk-restriction pair and showing that one of them produces trivial Bayesian updates.

2.1 Formalizing the asymmetry

Definition 1 (Risk-Restriction Pair)**.** *A* risk-restriction pair (R, Ξ) *consists of:*

- *A* risk claim $R = (a_j, (d_1, \ldots, d_k), \mathrm{Pathway}, p_R)$*, where* a_j *is the claiming agent,* $(d_1, \ldots, d_k)$ *is the capability tuple under concern,* $\mathrm{Pathway}$ *is the predicted exercise pathway [Lasser, 2026c, Definition 2] that would produce* $\mathrm{vol_P}$*-contraction, and* $p_R \in (0, 1]$ *is* a_j*'s claimed probability that* $\mathrm{Pathway}$ *activates if the tuple is exercised.*
- *A* restriction Ξ*: the preemptive restriction of a keystone capability* d_{j^*} *in the tuple (in the sense of [Lasser, 2026c, Proposition 1]), which prevents exercise of* $\mathrm{Pathway}$.

Definition 2 (Observation Regimes)**.** *Under a risk-restriction pair* (R, Ξ)*, the* observation regime *at time* t *is one of:*

- Restricted regime *(*Ξ *active): the capability tuple is restricted;* $\mathrm{Pathway}$ *cannot activate. The only observations available are indirect—behavioural signals from* a_j*, correlational evidence from related capabilities, and time-passage without incident (uninformative because the restriction is what prevents the test).*
- Exercise regime *(*Ξ *lifted): the capability tuple is exercised;* $\mathrm{Pathway}$ *may or may not activate. Direct observation of activation or non-activation is available.*

Proposition 1 (Asymmetric Evidence Under Restriction)**.** *Let* $p_R(t)$ *denote the posterior probability that risk claim* R *is valid at time* t*. Under the restricted regime:*

(a) If the pathway were to activate (counterfactual contraction observed under a hypothetical exercise), the standard Bayesian update yields

$$p_R(t+1) = p_R(t) \cdot \frac{\Pr(\textit{contraction} \mid R \textit{ valid})}{\Pr(\textit{contraction})}$$

with $\Pr(\textit{contraction} \mid R \textit{ valid}) > \Pr(\textit{contraction} \mid R \textit{ invalid})$*, which increases* p_R*.*

(b) Under the restricted regime, no exercise occurs, so

$$p_R(t+1) = p_R(t).$$

The posterior is unchanged because the restriction prevents the observation that would update it. The absence of contraction under restriction is consistent with both R *being valid (the restriction correctly prevented it) and* R *being invalid (there was nothing to prevent).*

Part (b) is the Wamura asymmetry*: successful prevention is observationally equivalent to unnecessary prevention.*

Proof sketch. Direct from the Bayesian update rule and the counterfactual structure. Under restriction, the likelihood ratio $\Pr(\text{no contraction} \mid R \text{ valid}, \Xi) / \Pr(\text{no contraction} \mid R \text{ invalid}, \Xi) = 1$ because both hypotheses predict no contraction when the pathway is blocked. Likelihood ratio 1 means the posterior equals the prior: no information is gained. The (a) direction is the standard Bayes-positive update; the asymmetry in (a) versus (b) is what produces the Wamura ratchet. ■

Remark 1 (Connection to substrate-level asymmetry)**.** *The verification-asymmetry definition of [Lasser, 2026a, Definition 2] is the substrate-level version of this result: a digital agent can demonstrate capability possession (positive claim) far more easily than an external observer can verify capability absence (negative claim). Proposition 1 is the risk-framework specialization: a risk claim can be confirmed by observing catastrophe but cannot be disconfirmed by observing non-catastrophe-under-restriction. The monotonicity result [Lasser, 2026a, Proposition 1] ensures the substrate-level asymmetry is non-decreasing in capability scale; Proposition 1 inherits the same monotone character for the risk-framework projection.*

2.2 The overcaution accumulation dynamic

Proposition 2 (Monotone Overcaution Under Max-Aggregation)**.** *Under max-aggregation of risk estimates [Lasser, 2026c] and neutral-prior initialization:*

(a) For any newly discovered capability tuple $(d_1, \ldots, d_k)$ with no exercise history, there exists a threshold $p_{\text{thresh}} > 0$ such that if any agent a_j assigns $p_R(0) > p_{\text{thresh}}$, the tuple is immediately restricted, by the preemptive restriction criterion of [Lasser, 2026c, Proposition 1].

(b) Once restricted, $p_R(t) = p_R(0)$ for all $t > 0$ under the restricted regime, by Proposition 1(b).

(c) The number of restricted capability tuples is monotonically non-decreasing over time: new tuples may be discovered and restricted, but no restricted tuple is unrestricted through the standard evidence-accumulation mechanism.

Proof sketch. Part (a) follows from the restriction criterion of [Lasser, 2026c, Proposition 1]: it compares $\mathcal{R}_j$ against the restriction cost, and for any tuple a sufficiently high p_R exceeds the threshold. Part (b) is Proposition 1(b). Part (c) is immediate from (b): no evidence accumulates to reduce p_R below p_{thresh}, and the existing framework contains no mechanism that generates exercise-regime observations for restricted tuples. ■

Corollary 1 (Asymptotic Lockup)**.** *In the limit of many capability discoveries, the fraction of the capability poset under active restriction grows monotonically, bounded below by the fraction of tuples for which at least one agent assigned $p_R(0) > p_{\text{thresh}}$. If agents are systematically overcautious ($p_R(0)$ exceeds the true risk on many tuples), the locked fraction can grow without bound relative to the truly risky fraction, producing vol_{P} contraction that the framework cannot distinguish from genuine safety.*

The corollary is the formal expression of the Fudai floodgate intuition taken to its pathological limit: if every agent assigns moderately high initial risk to every newly discovered capability, and no mechanism in the framework is able to differentiate genuine risk from overcaution, the poset contracts under restriction without any mechanism for reversal. This is precisely the degenerate phase boundary of Loop 3 in [Lasser, 2026d, Proposition 4]: the self-correction rate r_{S} is zero because no exogenous signal on the restricted dimension exists. The remainder of the paper constructs the missing mechanism.

3 Controlled Relaxation Protocol

The structural root of the Wamura problem is that evidence *against* a risk claim requires exercise of the restricted capability, which the restriction prevents. Controlled relaxation breaks this deadlock

by creating bounded, monitored exercise windows that generate genuine Bayesian evidence while bounding maximum damage.

3.1 The test object

Definition 3 (Controlled Relaxation Test)**.** *A* controlled relaxation test $T = (R, \Xi, S, \tau_{\text{test}}, \epsilon)$ *for a risk-restriction pair* (R, Ξ) *consists of:*

- *The risk-restriction pair* (R, Ξ) *being tested;*
- *A* test scope $S \subseteq P$*: a bounded sub-poset within which* Ξ *is temporarily lifted. The scope determines maximum* $\mathrm{vol}_{\mathrm{P}}$ *exposure (Section 4);*
- *A* test duration $\tau_{\text{test}} \in \mathbb{N}$*: the number of time steps for which* Ξ *is lifted within* S*; after* τ_{test} *steps,* Ξ *is reinstated regardless of outcome;*
- *A* damage tolerance $\epsilon > 0$*: the maximum acceptable* $\mathrm{vol}_{\mathrm{P}}$*-contraction from the test if the risk materializes.* ϵ *is set by the meta-policy (Section 7) or by the user as a safety bound.*

Remark 2 (CR tests the physical mechanism, not just the restriction flag)**.** *A controlled relaxation test does not ask "what happens if we remove the restriction?" It asks the sharper question "does the physical pathway that justified the restriction still activate when exercised?" The risk-restriction pair* (R, Ξ) *was adopted as a bet on a specific causal mechanism—the exercise pathway* $\mathrm{Pathway}$ *of [Lasser, 2026c, Definition 2]—operating from the capability tuple* $(d_1, \ldots, d_k)$ *to the predicted* $\mathrm{vol}_{\mathrm{P}}$*-contraction. Exercising the pathway inside* S *and observing its detection response is equivalent to probing the mechanism. The restriction may be stale not because the original risk analysis was wrong but because the underlying physical substrate has changed: new mitigation technology, evolved materials, different social context, or the intervening decades of technological progress that make the restriction's original triggering conditions no longer generic. The CR machinery inherits this framing from the pathway definition; Remark 6 names the alternative (structural resolution) for cases where the mechanism is mechanically infeasible in the current context.*

The test is conducted by actors whose integrity is appropriate to the domain; the formal construction is indifferent to the implementation (cryptographic commitment via [Lasser, 2026a], institutional oversight, legal accountability, or combinations thereof). Section 7 names the trust apparatus abstractly.

3.2 Protocol phases

Phase 1: Scope selection. Identify a sub-poset S such that

(i) the predicted pathway $\mathrm{Pathway}$ is exercisable within S (the prerequisite capabilities are available);

(ii) the maximum $\mathrm{vol}_{\mathrm{P}}$-contraction from any exercise pathway within S is bounded by ϵ (Theorem 1);

(iii) no other controlled relaxation test is currently running on a scope S' that overlaps with S (isolation, to prevent cascading test failures).

Phase 2: Relaxation. Within scope S, lift Ξ for τ_{test} time steps. During the test window:

- Monitor for the predicted contraction pathway using the multi-channel detection architecture of [Lasser, 2026f, Table 1]: capability gating, causal attribution, behavioural residuals, SPRT detection, correlational co-occurrence.
- Record all observations in the test log with timestamps and channel attributions (formalized in Section 5).
- If the detection channel reports $\mathrm{vol}_{\mathrm{P}}$-contraction within S exceeding ϵ at any step, immediately reinstate Ξ and terminate the test early (safety cutoff). The cutoff and the Bayesian-update observable O_n (Phase 3) read from the same noisy detector; a cutoff event is therefore a detection event in the sense of $O_n = 1$, not a separate exact-contraction channel.

Phase 3: Observation and Bayesian update. We separate the latent event (pathway activation) from the observed event (detection alarm). The model is a *simple binary hypothesis test* with fixed likelihoods throughout, in the sequential-analysis tradition of Wald [1947]; the closest methodological precedent is Bayesian adaptive trial design [Berry, 2006], which similarly conducts repeated tests under a bounded patient-risk budget and updates the posterior on each outcome. CR generalises this pattern from clinical trials to the broader class of long-timeline restrictions, replacing patient safety with the user-set damage tolerance ϵ and the Phase 1 cooperative-closure scope construction.

- $A_n \in \{0, 1\}$: indicator that the pathway activates during test n (latent; not directly observed);
- $O_n \in \{0, 1\}$: indicator that contraction is detected during test n (observed);
- H_0: the pathway is harmless—activation probability $\theta = 0$ (pathway never activates even when exercised);
- H_1: the pathway is dangerous—activation probability $\theta = \theta_1$ for a fixed $\theta_1 \in (0, 1]$ (pathway activates with known probability θ_1 when exercised).

The choice of θ_1 parametrizes the alternative hypothesis as a fixed point in $(0, 1]$. The conservative default is $\theta_1 = 1$ ("if the pathway is dangerous at all, it activates every time the capability is exercised"), which produces the strongest evidence per test. A more nuanced $\theta_1 < 1$ reduces evidence per test but may be more realistic for stochastically activating pathways. Under either choice, θ_1 is fixed across the test sequence and is *not* a free parameter in the likelihood.

Observation model. The detector is characterised by sensitivity and specificity:

$$\Pr(O_n = 1 \mid A_n = 1) = 1 - \beta(\tau_{\text{test}}) \qquad \text{(true-positive rate)},$$
$$\Pr(O_n = 1 \mid A_n = 0) = \alpha_{\text{fp}} \qquad \text{(false-positive rate)},$$

where $\beta(\tau_{\text{test}})$ is the false-negative rate over the test window (characterised in Section 5) and $\alpha_{\text{fp}} \in [0, 1)$ is the per-test false-positive rate of the aggregate detector.

Fixed per-hypothesis likelihoods. Marginalizing over A_n under each hypothesis gives, under H_1:

$$\Pr(O_n = 1 \mid H_1) = \theta_1(1 - \beta) + (1 - \theta_1)\alpha_{\text{fp}}, \tag{1}$$
$$\Pr(O_n = 0 \mid H_1) = \theta_1\beta + (1 - \theta_1)(1 - \alpha_{\text{fp}}),$$

and under H_0:

$$\Pr(O_n = 1 \mid H_0) = \alpha_{\text{fp}}, \tag{2}$$
$$\Pr(O_n = 0 \mid H_0) = 1 - \alpha_{\text{fp}}.$$

These are *fixed numbers* determined by $(\theta_1, \beta, \alpha_{\text{fp}})$, not functions of a free parameter. This is what makes the binary test well-defined.

Likelihood ratios. Define the per-test likelihood ratios

$$L_1 = \frac{\Pr(O_n = 1 \mid H_1)}{\Pr(O_n = 1 \mid H_0)} = \frac{\theta_1(1 - \beta) + (1 - \theta_1)\alpha_{\text{fp}}}{\alpha_{\text{fp}}}, \tag{3}$$
$$L_0 = \frac{\Pr(O_n = 0 \mid H_1)}{\Pr(O_n = 0 \mid H_0)} = \frac{\theta_1\beta + (1 - \theta_1)(1 - \alpha_{\text{fp}})}{1 - \alpha_{\text{fp}}}.$$

Under the identifiability assumption (Assumption 4 in Section 6) we will have $L_0 < 1$ and $L_1 > 1$; both are fixed scalars depending only on $(\theta_1, \beta, \alpha_{\text{fp}})$.

Posterior updates. For outcome $O_n = 1$ (contraction detected),

$$p_R(t^+) = \frac{p_R(t)\, L_1}{p_R(t)\, L_1 + (1 - p_R(t))}, \qquad \operatorname{logit} p_R(t^+) = \operatorname{logit} p_R(t) + \log L_1. \tag{4}$$

For outcome $O_n = 0$ (no contraction detected),

$$p_R(t^+) = \frac{p_R(t)\, L_0}{p_R(t)\, L_0 + (1 - p_R(t))}, \qquad \operatorname{logit} p_R(t^+) = \operatorname{logit} p_R(t) + \log L_0. \tag{5}$$

Because $L_0 < 1$ under Assumption 4, each no-contraction observation pushes the posterior toward H_0. This is what breaks the Wamura asymmetry.

Remark 3 (Contrast with the restricted regime)**.** *Under the restricted regime (Proposition 1), $A_n = 0$ with certainty (the restriction prevents activation), so both outcome likelihoods collapse to the common value and the effective $L_0 = L_1 = 1$: the posterior is unchanged. Under controlled relaxation, the capability is exercised (Assumption 8 in Section 6), activation can occur, and the detector can observe it—making both outcomes informative and breaking the ratchet.*

Remark 4 (Continuous extension)**.** *The binary formulation (H_0 versus H_1 with fixed θ_1) is sufficient for the operational question "should this restriction be maintained?" and avoids the dimensional mismatch between a binary posterior and a continuous activation rate. A continuous extension placing a prior $\pi(\theta)$ on $[0, 1]$ could estimate the exact activation rate but requires latent-state augmentation for the posterior update: the latent A_n makes the model non-conjugate with a Beta prior, unlike the standard Beta–Binomial. We defer this extension to future work; the binary formulation is what the Wamura problem requires.*

3.3 Connection to the phase boundary

Lasser [2026d] establishes that feedback loop 3 (over-calibrated restriction → discovery avoidance → structural ignorance) has a degenerate phase boundary when the self-correction rate $r_{\mathrm{S}} = 0$ for that loop [Lasser, 2026d, Proposition 4]. The self-correction rate is zero precisely because the Wamura asymmetry prevents evidence accumulation on the restricted dimension. Controlled relaxation injects discrete observation events at scheduled test times, producing a positive *time-averaged* exogenous correction rate $r_{\mathrm{S}}^{\mathrm{avg}} > 0$ on the restricted dimension.

The compositional detail is worth being explicit about. r_{S} as defined in [Lasser, 2026d] is $\inf_t \alpha(t) \cdot c_{\mathrm{S}}$, an infimum over the trajectory. Between CR test events the per-step rate $\alpha(t) \cdot c_{\mathrm{S}}$ on the restricted dimension is zero, so the inf-form r_{S} may remain zero even with periodic testing. The phase-boundary condition that actually closes the proof, however, is the uniform-contraction condition on *time-averaged* log-multiplicative factors [Lasser, 2026d, Theorem 1(a) and Eq. 30]; the inf-form r_{S} is upstream notational convenience.

CR delivers a positive time-average via the discrete log-odds drift of (4)–(5), which has nonzero expected magnitude per test under Assumption 4. The Wamura degeneracy that [Lasser, 2026d, Proposition 4] characterises is the case where the rate is identically zero across all t (no exogenous channel exists at all on the restricted dimension); CR resolves that degeneracy by creating exogenous channel events, satisfying the uniform-contraction condition in time-average without requiring the inf to be positive between scheduled tests. The restoring force that lifts Loop 3 off the phase-boundary degeneracy is therefore the time-averaged drift, not an instantaneous rate.

4 Scope-Bounding and Damage Guarantees

Section 3 treated the damage tolerance ϵ as a parameter; here we show that the scope construction makes ϵ an enforceable upper bound on the worst-case $\mathrm{vol}_{\mathrm{P}}$-contraction during and after a test, even when cooperative capabilities can propagate contraction outside the nominal scope.

4.1 Formalizing test scope

Definition 4 (Test Scope)**.** *A* test scope *S for risk claim $R = (a_j, (d_1, \ldots, d_k), \mathrm{Pathway}, p_R)$ is a sub-poset of P satisfying:*

(i) ***Pathway containment.*** $\mathrm{Pathway}$ *is exercisable within S: all prerequisite capabilities are present in S, and all intermediate poset transitions along* $\mathrm{Pathway}$ *remain within S.*

(ii) ***Forward-reachable cooperative closure.*** *For every cooperative capability c in the full poset P that involves at least one capability in S and at least one capability in $P \setminus S$, the maximum $\mathrm{vol}_{\mathrm{P}}$-contraction along any pathway within S that modifies c is included in* $\mathrm{max_contraction}(S)$*. Furthermore, if modification of c can causally enable (unlock) an exercise pathway in $P \setminus S$—that is, there exists a pathway $\mathrm{Pathway}_{\mathrm{ext}}$ in $P \setminus S$ whose prerequisites include a state change in c—then the expected contraction from $\mathrm{Pathway}_{\mathrm{ext}}$, weighted by its activation probability, is also included in* $\mathrm{max_contraction}(S)$*. This transitive-closure condition prevents a test within S from triggering a cascade outside S through cross-boundary cooperative capabilities.*

(iii) ***Irreversibility bound.*** *For each exercise pathway within S, the fraction of the pathway's $\mathrm{vol}_{\mathrm{P}}$-contraction that is irreversible (cannot be recovered by reinstating Ξ) is bounded by a known constant $\delta_{\mathrm{irrev}}(\mathrm{Pathway}) \leq \epsilon$. Pathways with $\delta_{\mathrm{irrev}} > \epsilon$ are excluded from the scope.*

(iv) ***Scope minimality.*** *S is inclusion-minimal subject to (i)–(iii): no proper sub-poset of S satisfies all three conditions.*

Remark 5 (Constructibility of the scope)**.** *Condition (i) is computable from the exercise-pathway definition of [Lasser, 2026c, Definition 2]: enumerate the prerequisite set and all intermediate states. Condition (ii) requires computing the cooperative capabilities that span S and $P \setminus S$—the* cooperative surface *of S—and then forward-reachability analysis: for each boundary cooperative, enumerate external pathways whose prerequisites include a state change in that cooperative. This is more expensive than simple pathway enumeration (exponential in the depth of the forward-reachability chain) but is necessary to prevent cascading damage through cooperative-capability spillover. In practice the forward-reachability depth is bounded by the poset diameter and most cooperative capabilities have low fan-out. Condition (iii) requires a pathway-level reversibility analysis; most capability restrictions are reversible (reinstatement restores the prior state), but information-disclosure and trust-degradation pathways may have irreversible components.*

4.2 The damage-bound theorem

Theorem 1 (Damage Bound for Controlled Relaxation)**.** *Let $T = (R, \Xi, S, \tau_{\mathrm{test}}, \epsilon)$ be a controlled relaxation test with test scope S satisfying Definition 4. Then:*

(a) ***Maximum instantaneous damage.*** *At any step $t' \in \{t, t+1, \ldots, t+\tau_{\mathrm{test}}\}$ during the test,*

$$\mathrm{vol}_{\mathrm{P}}(G_{t'}) \geq \mathrm{vol}_{\mathrm{P}}(G_t) - \mathrm{max_contraction}(S),$$

where $\mathrm{max_contraction}(S)$ is the maximum $\mathrm{vol}_{\mathrm{P}}$-contraction achievable by any exercise pathway within S. Under the scope-selection constraint $\mathrm{max_contraction}(S) \leq \epsilon$, the instantaneous damage is bounded by ϵ.

(b) ***Partial reversibility under reinstatement.*** *If the safety cutoff triggers at some step $t' < t + \tau_{\mathrm{test}}$ (the detector reports contraction within S exceeding ϵ; see Section 3 Phase 2 for the single-detector convention), reinstating Ξ at $t'+1$ gives*

$$\mathrm{vol}_{\mathrm{P}}(G_{t'+1}) \geq \mathrm{vol}_{\mathrm{P}}(G_t) - \epsilon - \delta_{\mathrm{irrev}}^{\max},$$

where $\delta_{\mathrm{irrev}}^{\max} = \max_{\mathrm{Pathway}' \subseteq S} \delta_{\mathrm{irrev}}(\mathrm{Pathway}')$ is the maximum irreversible component across all pathways in the scope (Definition 4(iii)). By the scope construction $\delta_{\mathrm{irrev}}^{\max} \leq \epsilon$, so total post-reinstatement damage is bounded by 2ϵ.

(c) ***Scope isolation under cooperative closure.*** *If no other test is running on an overlapping scope (Definition 3, Phase 1, condition (iii)) and S satisfies cooperative closure (Definition 4(ii)), then a cooperative-closed S exists (Lemma 1 gives a constructive enumeration of the cooperative surface and its forward-reachability closure), the cross-boundary contribution to contraction is finite and enumerable, and the damage from test T that propagates to $P \setminus S$ is bounded by the cooperative-surface terms that Definition 4(ii) requires to be included in $\mathrm{max_contraction}(S)$. Formally: for any capability $d \notin S$ that is not part of any cooperative capability spanning S and $P \setminus S$, the exercise probabilities of d's pathways are unchanged during the test. The substantive content of part (c) is therefore the existence and tractability of the closure; the containment bound follows by construction from Definition 4(ii).*

Proof sketch. Part (a) follows from the pathway-enumeration construction of $\mathrm{max_contraction}$ under finite pathway length, and the scope-selection constraint. Part (b) decomposes the contraction at the cutoff step into its reversible and irreversible components; the reversible component is restored by reinstating Ξ, the irreversible component is bounded above by ϵ under Definition 4(iii). Part (c) is the direct consequence of cooperative closure: cross-boundary cooperative terms are accounted for in $\mathrm{max_contraction}(S)$ by construction, and all other boundary capabilities are untouched by the test. Full derivation is in Appendix A, Section A.3. ■

Remark 6 (Scope existence as a scheduling precondition)**.** *Theorem 1 is stated conditionally on the existence of a scope satisfying Definition 4. For posets with dense cross-boundary cooperative structure, the cooperative-closure requirement may force $S = P$, in which case* $\mathrm{max_contraction}(P) \leq \epsilon$ *may fail and no test is schedulable. The feasibility condition is enforced by the meta-policy (Section 7, Definition 6, Phase 1); Theorem 1 applies only when the meta-policy reports a feasible scope. When no feasible scope exists, the restriction persists and the Wamura problem is not resolved for that specific risk claim by controlled relaxation; alternative mechanisms (structural discovery via [Lasser, 2026c, Proposition 2], or scope-refinement that trades coverage for feasibility) must apply.*

Remark 7 (Why not $\delta_{\mathrm{irrev}} \leq 0$?)**.** *The preemptive restriction criterion [Lasser, 2026c, Proposition 1] is an ex-ante cost-benefit comparison justifying the initial restriction; it does not guarantee that reinstatement after partial pathway execution fully recovers the pre-test state. Irreversible state changes during the test (information disclosure to adversaries, trust degradation, poset structural changes not reversed by restriction alone) create a gap between the pre-test* $\mathrm{vol_P}$ *and the post-reinstatement* $\mathrm{vol_P}$*. Definition 4(iii) bounds this gap by requiring the scope to exclude pathways with large irreversible components.*

Lemma 1 (Computable max_contraction)**.** *For a finite sub-poset S with bounded pathway length L and bounded cooperative-surface forward-reachability depth L',* $\mathrm{max_contraction}(S)$ *is computable in $O(|S|^{L+1} + |P|^{L'})$ time by exhaustive pathway enumeration together with enumeration of cross-boundary cooperative chains. When the simplifying assumption $L' \leq L$ holds (cross-boundary cascades cannot chain more than pathway-length-many cooperative hops), the bound simplifies to $O(|P|^{L+1})$.*

Proof sketch. Pathway length is bounded in [Lasser, 2026c, Definition 2]. The number of internal pathways is at most $|S|^L$ (each step selects a capability in S); each pathway contraction is computable from the axioms M1–M6 [Lasser, 2026e, Proposition 1] in $O(|S|)$ time. The cooperative-surface terms from Definition 4(ii) require enumerating cross-boundary cooperative chains to depth L', costing $O(|P|^{L'})$ in the worst case. Take the maximum across internal pathways and add cooperative-surface terms. ■

Remark 8 (Computational tractability)**.** *The bound is exponential in pathway length and forward-reachability depth but polynomial in scope size for fixed L, L'. Practical deployment requires either (a) short pathway restrictions ($L \leq 3$, say), (b) approximate bounds using the concentration-risk formula of [Lasser, 2026c, Definition 3] as an upper bound on* $\mathrm{max_contraction}$*, or (c) structural decomposition of S into independent sub-scopes. This is an engineering consideration, not a theoretical limitation—the bound exists and is computable; the question is computational cost.*

Corollary 2 (Damage bound under conservative over-approximation)**.** *Let* $\mathrm{max_contraction}^\star(S) \geq \mathrm{max_contraction}(S)$ *be any over-approximation of the maximum scope contraction. A scope-selection rule that requires* $\mathrm{max_contraction}^\star(S) \leq \epsilon$ *continues to satisfy the damage bound of Theorem 1: worst-case* $\mathrm{vol_P}$*-contraction during the test is* $\leq \mathrm{max_contraction}(S) \leq \mathrm{max_contraction}^\star(S) \leq \epsilon$*, and the post-cutoff bound 2ϵ inherits unchanged. Cooperative-closure enumeration cost is therefore tradeable against scope conservatism: deployers may accept a narrower S chosen under a cheaper-to-compute over-approximation of the cooperative surface and still preserve the safety guarantee, at the cost of feasibility (some otherwise-feasible scopes will be rejected).*

Proof sketch. Direct. By assumption $\mathrm{max_contraction}(S) \leq \mathrm{max_contraction}^\star(S) \leq \epsilon$, so Theorem 1(a)'s inequality $\mathrm{vol_P}(G_{t'}) \geq \mathrm{vol_P}(G_t) - \mathrm{max_contraction}(S) \geq \mathrm{vol_P}(G_t) - \epsilon$ holds. Part (b) inherits because the irreversibility bound $\delta^{\max}_{\mathrm{irrev}} \leq \epsilon$ of Definition 4(iii) is a pathway-level condition unaffected by the maxcon over-approximation. Part (c) inherits because cooperative-surface contractions are included inside $\mathrm{max_contraction}$ by construction (Definition 4(ii)) and $\mathrm{max_contraction} \leq \mathrm{max_contraction}^\star \leq \epsilon$. ■

5 Detection Sensitivity and the Bayesian Update

The Bayesian update in Section 3 depends on the detector's false-negative rate $\beta(\tau_{\mathrm{test}})$ over the test window. This section characterizes β in terms of the multi-channel detection architecture of [Lasser, 2026f, Table 1] and records the per-test evidence strength that follows.

5.1 The detection model

Within a single test, $A_n = 1$ means the pathway activates at some point during the τ_{test}-step window; $O_n = 1$ means the detector records contraction during that window. The detection sensitivity β mediates between them: $\beta = \Pr(O_n = 0 \mid A_n = 1)$.

Assumption 1 (Activation Duration)**.** *When the pathway activates during a test ($A_n = 1$), the activation persists for k_{act} consecutive steps, where k_{act} is either (a) a fixed parameter of the pathway (determined by [Lasser, 2026c, Definition 2]'s exercise-pathway structure), or (b) drawn from a distribution K with known or bounded support. The per-test false-negative rate β depends on k_{act}: the detector has $\min(k_{\text{act}}, \tau_{\text{test}})$ steps to detect activation. For the default analysis we assume $k_{\text{act}} \geq \tau_{\text{test}}$ (persistent activation); the transient case $k_{\text{act}} < \tau_{\text{test}}$ is handled as a sensitivity analysis below.*

Definition 5 (Detection Sensitivity)**.** *For a test $T = (R, \Xi, S, \tau_{\text{test}}, \epsilon)$ monitoring pathway $\text{Pathway} = (\pi_1, \ldots, \pi_L)$, the per-step detection probability on channel c_i is*

$$\delta_i = \Pr(c_i \textit{ detects activation at } t' \mid \text{Pathway } \textit{activates at } t').$$

The aggregate per-step detection probability across all active channels is

$$\delta_{\text{agg}} = 1 - \prod_{c_i \in \mathcal{O}(P_t) \cap \mathcal{C}_S} (1 - \delta_i), \tag{6}$$

where $\mathcal{C}_S$ is the set of channels that cover the scope S.

Assumption 2 (Cross-Channel Independence)**.** *Detection events on distinct channels are conditionally independent given the latent activation state A_n. This is justified for cross-substrate channels by channel isolation [Lasser, 2026a, Proposition 5] (unconditional), and for same-substrate channels under the joint-factorization assumption of [Lasser, 2026a, Proposition 6] (conditional on committed verification). When Assumption 2 fails (correlated same-substrate channels without committed verification), the product formula* (6) *is optimistic; a conservative fallback uses only the cross-substrate channels,*

$$\delta_{\text{agg}}^{\text{cons}} = 1 - \prod_{c_i \in \mathcal{O}(P_t) \cap \mathcal{C}_S:\ \text{cross-substrate}} (1 - \delta_i). \tag{7}$$

Assumption 3 (Temporal Independence or Mixing)**.** *Conditional on the latent activation state A_n, detection outcomes at distinct time steps within a test window are either (a) independent (memoryless detectors) or (b) satisfy a mixing condition with mixing time $\tau_{\text{mix}} \ll \tau_{\text{test}}$. Without Assumption 3, persistent detector outages or common-cause failures can produce correlated misses across time steps, making β decay slower than exponential. Under (b), the effective number of independent detection opportunities in τ_{test} steps is $\tau_{\text{test}}/\tau_{\text{mix}}$ rather than τ_{test}.*

Under Assumptions 1–3 with persistent activation, the window-level false-negative rate satisfies

$$\beta(\tau_{\text{test}}) \leq (1 - \delta_{\text{agg}})^{\tau_{\text{test}}/\tau_{\text{mix}}}. \tag{8}$$

Under the memoryless case (3(a)), $\tau_{\text{mix}} = 1$ and $\beta(\tau_{\text{test}}) = (1 - \delta_{\text{agg}})^{\tau_{\text{test}}}$. For transient pathways with activation duration $k < \tau_{\text{test}}$, replace τ_{test} by $\min(k, \tau_{\text{test}})$ throughout.

5.2 Detection-sensitivity bounds

Proposition 3 (Detection Sensitivity Bounds)**.** *Under the multi-channel architecture of [Lasser, 2026f, Table 1] and Assumptions 1–3:*

(a) $\beta(\tau_{\text{test}})$ is monotonically decreasing in τ_{test}: longer tests have lower false-negative rates.

(b) For any $\epsilon_\beta \in (0, 1)$, the minimum test duration required to achieve $\beta(\tau_{\text{test}}) \leq \epsilon_\beta$ is

$$\tau_{\min} = \left\lceil \frac{\log \epsilon_\beta}{\log(1 - \delta_{\text{agg}})} \right\rceil, \tag{9}$$

rescaled by τ_{mix} under the mixing case of Assumption 3(b).

(c) *If at least one cross-substrate channel covers S with detection probability $\delta_{\text{cross}} > 0$, then even without Assumption 2 for same-substrate channels,*

$$\beta(\tau_{\text{test}}) \leq (1 - \delta_{\text{cross}})^{\tau_{\text{test}}/\tau_{\text{mix}}}, \tag{10}$$

which converges to zero exponentially in τ_{test} at rate $1/\tau_{\text{mix}}$.

Proof sketch. Part (a) is immediate from (8): adding more independent detection opportunities monotonically decreases the product. Part (b) inverts the bound in (8). Part (c) uses that cross-substrate channels satisfy Assumption 2 unconditionally by [Lasser, 2026a, Proposition 5], so the product formula is exact for them; temporal independence follows from Assumption 3(a) or (b). ■

Part (c) is the substrate-diversity dividend for detection: cross-substrate channels provide independent detection opportunities that same-substrate channels may not (due to correlated failures), and their isolation is unconditional.

Remark 9 (Cross-substrate-only rate recovery)**.** *The interaction between Assumptions 2 (cross-channel independence) and 3 (temporal independence) is subtle. A common-cause failure mode that violates 2 for same-substrate channels (e.g., a shared software dependency that miscalibrates an entire silicon detector cohort) typically also violates 3 for the same-substrate temporal stream (the failure persists across time). The actually-dangerous regime is therefore* joint *same-substrate failure across channels and across time. The conservative bound $\delta_{\text{agg}}^{\text{cons}}$ of* (7) *protects against both simultaneously: cross-substrate channels, by [Lasser, 2026a, Proposition 5]'s unconditional channel-isolation, satisfy 2* and *3(a) without further assumption, because the temporal failure of a biological observer is independent of the temporal failure of a silicon observer. Under $\delta_{\text{agg}}^{\text{cons}} > 0$, Theorem 2's KL rate is recovered up to a factor that depends only on the ratio of cross-substrate to all-channel detection probability: $\text{KL}^{\text{cons}}/\text{KL}^{\text{full}} \geq \delta_{\text{agg}}^{\text{cons}}/\delta_{\text{agg}}$ in the regime $\delta_{\text{agg}} \ll 1$, with equality in the limit of a single dominant channel. Convergence is preserved unconditionally; the constant factor degradation is the cost of the joint failure mode.*

5.3 Composition rule for transient pathways

Assumption 1 treats persistent activation as the default ($k_{\text{act}} \geq \tau_{\text{test}}$). For transient pathways with activation duration $k < \tau_{\text{test}}$, the window false-negative rate is $\beta(\tau_{\text{test}}) = (1 - \delta_{\text{agg}})^{\min(k, \tau_{\text{test}})}$ under Assumption 3(a), or scaled by τ_{mix} under Assumption 3(b). When $k = 1$ (single-step activation) the formula gives $\beta = 1 - \delta_{\text{agg}}$, which may be non-negligible. An operational composition rule fixes the design parameter.

Proposition 4 (Detection design rule for transient pathways)**.** *For a target window false-negative rate ϵ_β and a known (or upper-bounded) pathway duration k:*

- *Under Assumption 3(a) (memoryless detectors), the per-step detection probability δ_{agg} during the test must satisfy $(1 - \delta_{\text{agg}})^k \leq \epsilon_\beta$, i.e., $\delta_{\text{agg}} \geq 1 - \epsilon_\beta^{1/k}$. For $k = 1$ this collapses to $\delta_{\text{agg}} \geq 1 - \epsilon_\beta$ (the single-step detector must itself meet the target); for larger k the requirement softens proportionally.*
- *Under Assumption 3(b) (mixing with mixing time τ_{mix}), the effective number of independent detection opportunities in k steps is k/τ_{mix}, and the rule strengthens to $\delta_{\text{agg}} \geq 1 - \epsilon_\beta^{\tau_{\text{mix}}/k}$; recover memoryless as $\tau_{\text{mix}} \to 1$.*

Implementations that cannot raise δ_{agg} at fixed τ_{test} must either extend the test window (use the persistent-activation form with longer τ_{test}) or raise the detector sampling rate within the window (effectively shrinking τ_{mix}). The choice between these is implementation-defined; the composition rule sets the constraint each must satisfy.

5.4 Evidence strength of a single test

Proposition 5 (Evidence Strength of Controlled Relaxation)**.** *After a controlled relaxation test T with outcome $O_n \in \{0, 1\}$, using the binary model of Section 3 (H_0: $\theta = 0$ vs. H_1: $\theta = \theta_1$ with fixed likelihoods):*

(a) ***Evidence for*** R ***(***$O_n = 1$***, contraction detected).***

$$\Delta \operatorname{logit} = \log L_1 = \log \frac{\theta_1(1-\beta) + (1-\theta_1)\alpha_{\text{fp}}}{\alpha_{\text{fp}}}. \tag{11}$$

This is a positive constant under Assumption 4 ($L_1 > 1$*). For the common case* $\theta_1 = 1$*:* $\Delta \operatorname{logit} = \log((1-\beta)/\alpha_{\text{fp}})$.

(b) ***Evidence against*** R ***(***$O_n = 0$***, no contraction detected).***

$$\Delta \operatorname{logit} = \log L_0 = \log \frac{\theta_1\beta + (1-\theta_1)(1-\alpha_{\text{fp}})}{1-\alpha_{\text{fp}}}. \tag{12}$$

This is a negative constant under Assumption 4 ($L_0 < 1$*). For* $\theta_1 = 1$*:* $\Delta \operatorname{logit} = \log(\beta/(1-\alpha_{\text{fp}}))$*. The denominator* $(1-\alpha_{\text{fp}})$ *is material when* α_{fp} *is non-negligible and was missing in a naive derivation that assumed* $\alpha_{\text{fp}} = 0$.

(c) ***Comparison with the restricted regime.*** *Under the restricted regime (Proposition 1), the capability is not exercised, so both outcomes have likelihood ratio* 1 *(no update). Under controlled relaxation the capability is exercised (Assumption 8), so both* L_0 *and* L_1 *are non-unit and both outcomes are informative.*

(d) ***Effect of*** θ_1***.*** *The alternative hypothesis* θ_1 *affects evidence strength:*

- $\theta_1 = 1$*:* $L_1 = (1-\beta)/\alpha_{\text{fp}}$, $L_0 = \beta/(1-\alpha_{\text{fp}})$*. Maximum evidence per test.*
- $\theta_1 < 1$*:* L_1 *decreases and* $|\log L_0|$ *decreases. Less evidence per test, but the alternative may be more realistic for stochastically activating pathways.*

The choice of θ_1 *is a modelling decision that does not affect convergence (Theorem 2 holds for any* $\theta_1 \in (0, 1]$ *under Assumption 4), only the rate.*

Proof sketch. Direct from the fixed likelihood ratios (3). The log-odds form follows from (4) and (5): $\operatorname{logit} p_R(t^+) = \operatorname{logit} p_R(t) + \log(\Pr(O_n \mid H_1)/\Pr(O_n \mid H_0))$. ■

Remark 10 (Integration with log-odds aggregation)**.** *The log-odds evidence from Proposition 5 feeds directly into the log-odds risk aggregation of [Lasser, 2026f, Definition 5] as an additional evidence channel. The controlled-relaxation test produces evidence with the same structure as an individual agent's risk assessment—a log-odds contribution weighted by trust. The trust weight on this evidence is* 1 *(the observation is direct, not mediated by an agent's report), making test outcomes the highest-trust evidence the aggregation can receive. This is by design: direct observation under controlled conditions is the gold standard for empirical risk assessment.*

6 Convergence of Risk Estimates

This section states and proves the central convergence result: under repeated controlled relaxation with fixed parameters, the posterior p_R converges almost surely to the correct hypothesis. The result is stated in the simple binary hypothesis framework of Section 3.

6.1 Formulation

The convergence result is stated in the *binary hypothesis* framework (H_0: $\theta = 0$; H_1: $\theta = \theta_1 > 0$ for a fixed θ_1). This is the dimensionally consistent formulation: p_R is the posterior probability of H_1 and converges to 0 or 1, not to a fractional activation rate.

The continuous estimation problem (inferring the exact value of $\theta^{\text{true}} \in [0, 1]$) is strictly harder: it requires a continuous prior $\pi(\theta)$ and produces a posterior density, not a point estimate. The binary formulation suffices for the operational question "should this restriction be maintained?" and is what the Wamura problem requires.

6.2 Assumptions

Assumption 4 (Identifiable Binary Model)**.** *The observation model of Section 3 discriminates H_0 from H_1: the likelihood ratios satisfy $L_0 < 1$ and $L_1 > 1$. Equivalently, $\theta_1(1-\beta) > \theta_1\alpha_{\text{fp}}$, which holds whenever $(1-\beta) > \alpha_{\text{fp}}$ (the detector has nontrivial power), satisfied whenever at least one channel with nonzero sensitivity covers the scope (Proposition 3(c)).*

Assumption 5 (Correct Likelihood Specification)**.** *The observation model used in the Bayesian update (sensitivity $1-\beta$, specificity $1-\alpha_{\text{fp}}$, and fixed alternative θ_1) matches the true data-generating process. Misspecification (e.g., the actual false-positive rate differs from the assumed α_{fp}) produces convergence to the wrong hypothesis (Discussion, Section 11).*

Assumption 6 (Inter-Test Independence)**.** *Test outcomes $\{O_n\}$ are conditionally independent given the true state of the world (H_0 or H_1). This requires: (a) no adaptation by the tested agent between tests (the agent does not learn to evade detection based on prior results); (b) the detector's false-positive and false-negative rates are stable across tests (no detector degradation); and (c) the poset state is approximately stationary between tests (the scope S remains valid). Failure of (a) is the adversarial case (Open Question 3, Section 11).*

Assumption 7 (Prior Support)**.** *The prior $p_R(0) \in (0,1)$: neither hypothesis is assigned probability zero a priori. This is a standard requirement for Bayesian consistency: if $p_R(0) = 0$ (resp. $= 1$), no amount of evidence can move the posterior. In practice, the neutral-prior initialization of [Lasser, 2026c] assigns $p_R(0)$ in a neighbourhood of 0.5.*

Assumption 8 (Capability Exercise During Test)**.** *During the test window, the capability tuple $(d_1, \ldots, d_k)$ is actually exercised—the conditions for pathway activation are present, not merely that the restriction is lifted. Without Assumption 8, $O_n = 0$ is ambiguous between "the pathway is safe" and "nobody tried." In practice the scope-selection phase (Phase 1) enforces Assumption 8: the scope is constructed to include prerequisite capabilities, and the protocol actively invokes the tuple rather than waiting. When the assumption cannot be guaranteed (e.g., the pathway requires a specific environmental trigger), the evidential weight of $O_n = 0$ must be discounted by the probability p_{ex} that the capability was actually exercised: $\Pr(O_n = 0 \mid H_1) = p_{\text{ex}}[\theta_1\beta + (1-\theta_1)(1-\alpha_{\text{fp}})] + (1-p_{\text{ex}})(1-\alpha_{\text{fp}})$.*

6.3 Main result

Theorem 2 (Convergence Under Repeated Controlled Relaxation)**.** *Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of controlled relaxation tests for a risk-restriction pair (R, Ξ) with fixed parameters $(S, \tau_{\text{test}}, \epsilon)$, under the binary formulation H_0 ($\theta = 0$) versus H_1 ($\theta = \theta_1$). Under Assumptions 4–8 with $p_{\text{ex}} = 1$ (the full-exercise regime of Assumption 8, enforced by scope construction in Phase 1):*

(a) ***Almost-sure convergence.*** *Under the true hypothesis H_j ($j \in \{0,1\}$), the posterior converges almost surely to the correct value:*

$$H_0 \text{ true}: \quad p_R(t_n) \xrightarrow{\text{a.s.}} 0; \qquad H_1 \text{ true}: \quad p_R(t_n) \xrightarrow{\text{a.s.}} 1.$$

(b) ***Rate under H_0 (pathway harmless).*** *Every test produces $O_n = 0$ with probability $1-\alpha_{\text{fp}}$ and $O_n = 1$ with probability α_{fp}. Conditional on a run of n tests with all $O_n = 0$,*

$$\operatorname{logit} p_R(t_n) = \operatorname{logit} p_R(0) + n \log L_0,$$

so $p_R \to 0$ geometrically at rate $|\log L_0|$ per test.

(c) ***Rate under H_1 (pathway dangerous).*** *Each test produces $O_n = 1$ with probability $\Pr(O = 1 \mid H_1) = \theta_1(1-\beta) + (1-\theta_1)\alpha_{\text{fp}}$. The log-odds update on $O_n = 1$ is $\log L_1 > 0$, which is large when $(1-\beta) \gg \alpha_{\text{fp}}$. After a single $O = 1$ observation, the posterior is exactly $p_R(t_1) = p_R(0)L_1/(p_R(0)L_1 + (1-p_R(0)))$, so a single detection suffices to push p_R above $1-\delta$ for any $\delta > \delta_{\min}$, where $\delta_{\min} = (1-p_R(0))/(p_R(0)L_1 + (1-p_R(0)))$. For θ_1 close to 1, small β, and modest $p_R(0)$, $\delta_{\min}$ is small (e.g., $p_R(0) = 0.5$, $L_1 = 100$ gives $\delta_{\min} \approx 0.01$); for tighter δ requirements, the expected number of tests grows as $O\big(1/[\theta_1(1-\beta)]\big)$.*

(d) ***General convergence rate (KL form).*** *The log-posterior converges at a rate given by the KL divergence between the true observation distribution and the alternative:*

$$\begin{aligned} H_0 \textit{ true} &: \mathrm{KL}(\Pr(O \mid H_0) \,\|\, \Pr(O \mid H_1)) \\ &= (1-\alpha_{\mathrm{fp}})\log(1/L_0) + \alpha_{\mathrm{fp}}\log(1/L_1), \\ H_1 \textit{ true} &: \mathrm{KL}(\Pr(O \mid H_1) \,\|\, \Pr(O \mid H_0)) \\ &= \Pr(O=1 \mid H_1)\log L_1 + \Pr(O=0 \mid H_1)\log L_0. \end{aligned}$$

Both rates are strictly positive under Assumption 4, confirming exponential convergence in both directions.

Proof sketch. Under Assumptions 4–7, the log-likelihood ratio $\Lambda_n = \sum_{i=1}^{n} \log(\Pr(O_i \mid H_1)/\Pr(O_i \mid H_0))$ is a sum of i.i.d. bounded random variables (bounded because L_0, L_1 are fixed positive constants and $O_i \in \{0,1\}$). Under the true hypothesis H_j, the expected increment is $\pm\mathrm{KL}(\Pr(O \mid H_j) \,\|\, \Pr(O \mid H_{1-j}))$, strictly nonzero by Assumption 4 and the positivity of KL on non-identical distributions. The strong law of large numbers [Cover and Thomas, 2006, Ch. 6] (or the martingale convergence theorem of Doob [1953], which gives the same conclusion under weaker independence assumptions) gives $\Lambda_n/n \to \pm\mathrm{KL}$ a.s., hence $\Lambda_n \to \pm\infty$ a.s., hence $p_R(t_n) \to 1$ (under H_1) or $\to 0$ (under H_0). Part (b) follows from direct summation under the run of n all-negative tests; Part (c) from the geometric distribution of first detection; Part (d) from the definition of KL applied to the binary outcome distribution (Chernoff's result in Chernoff [1952] gives the associated large-deviation exponent). Full derivation in Appendix A.6. ■

Remark 11 (Dependence on Theorem 1)**.** *Theorem 2 is stated under the operating assumptions of Theorem 1: each test T_n is assumed to run with a feasible scope. Because the Phase 2 safety-cutoff reads from the same noisy detector that supplies O_n (Section 3), a cutoff event* is *an $O_n = 1$ observation with likelihood ratio L_1—the cutoff fires precisely because the detector alarmed, so the cutoff is evidence for H_1, not a distinct event outside the observation model. The window-truncation from early cutoff does mean the effective β for that particular test is smaller than $\beta(\tau_{\mathrm{test}})$ (the miss probability within $k \le \tau_{\mathrm{test}}$ steps is $(1-\delta_{\mathrm{agg}})^k$); using the full-window L_1 for a truncated test is therefore slightly conservative (understates the evidence) but remains a valid per-test lower bound on $|\Delta_{\log\,\mathrm{odds}}|$. The irreversible-damage component from Theorem 1(b) affects the realised* $\mathrm{vol}_{\mathrm{P}}$ *loss, not the Bayesian update. If no feasible scope exists (Remark 6), the sequence $\{T_n\}$ is not realisable and Theorem 2 does not apply.*

6.4 Finite-sample guarantees

Corollary 3 (Minimum Tests to Clear a False Claim)**.** *Under the operating assumptions of Theorem 2 (full-exercise regime of Assumption 8, $p_{\mathrm{ex}} = 1$), to reduce a false risk claim from prior $p_R(0)$ to posterior $p_R \le p_{\mathrm{clear}}$ when H_0 is true and all n tests produce $O_n = 0$ (no spurious false positives), the required number of tests satisfies*

$$N_{\mathrm{clear}} \ge \left\lceil \frac{\log\big(p_{\mathrm{clear}}(1-p_R(0))/(p_R(0)(1-p_{\mathrm{clear}}))\big)}{\log L_0} \right\rceil, \tag{13}$$

where $L_0 = [\theta_1\beta + (1-\theta_1)(1-\alpha_{\mathrm{fp}})]/(1-\alpha_{\mathrm{fp}})$. Note $L_0 < 1$ by Assumption 4, so the right-hand side is positive. When $\alpha_{\mathrm{fp}} \ll 1$, $L_0 \approx \theta_1\beta + (1-\theta_1)$ and the formula reduces approximately to

$$N_{\mathrm{clear}} \approx \left\lceil \frac{\operatorname{logit} p_{\mathrm{clear}} - \operatorname{logit} p_R(0)}{\log(\theta_1\beta + 1 - \theta_1)} \right\rceil.$$

Proof sketch. From (5) applied n times, $\operatorname{logit} p_R(t_n) = \operatorname{logit} p_R(0) + n\log L_0$. Setting $\operatorname{logit} p_R(t_n) \le \operatorname{logit} p_{\mathrm{clear}}$ and solving for n yields the bound. ■

Remark 12 (Concrete numbers)**.** *For $\theta_1 = 1$, $\beta = 0.1$, $\alpha_{\mathrm{fp}} = 0.01$, $p_R(0) = 0.5$, and $p_{\mathrm{clear}} = 0.01$: $L_0 = (1\cdot 0.1 + 0\cdot 0.99)/0.99 = 0.101$, and*

$$N_{\mathrm{clear}} = \left\lceil \frac{\log\big(0.01\cdot 0.5/(0.5\cdot 0.99)\big)}{\log 0.101} \right\rceil = \lceil 2.006 \rceil = 3 \textit{ tests}.$$

Three controlled relaxation tests suffice to clear a false risk claim from a neutral prior to 1% residual risk, given 90% detection sensitivity and 1% false-positive rate. The correction from including α_{fp} adds approximately one test compared with the naive β^N estimate. This illustrates the power of direct observation: the verification asymmetry that makes the Wamura problem seem intractable is broken by a small number of well-designed tests.

Sensitivity of N_{clear} to the model parameters (computed from (13) *with $\theta_1 = 1$ throughout):*

β	α_{fp}	$p_R(0)$	p_{clear}	L_0	N_{clear}
0.10	*0.01*	*0.50*	*0.01*	*0.101*	*3*
0.10	*0.01*	*0.70*	*0.01*	*0.101*	*3*
0.10	*0.01*	*0.90*	*0.01*	*0.101*	*3*
0.01	*0.01*	*0.50*	*0.01*	*0.0101*	*1*
0.30	*0.01*	*0.50*	*0.01*	*0.303*	*4*
0.50	*0.01*	*0.50*	*0.01*	*0.505*	*7*
0.10	*0.05*	*0.50*	*0.01*	*0.105*	*3*
0.10	*0.20*	*0.50*	*0.01*	*0.125*	*3*

The detection-sensitivity parameter β dominates the count; the false-positive rate α_{fp} contributes weakly until $\alpha_{\mathrm{fp}} \to (1-\beta)$, where the likelihood ratio L_0 approaches 1 and convergence stalls. For $\theta_1 < 1$ (realistic activation rates below the maximum), L_0 increases toward 1 proportionally to $1-\theta_1$, and N_{clear} grows accordingly.

Remark 13 (Robustness to p_{ex} misestimation)**.** *The convergence rate of Theorem 2 is stated under the full-exercise regime of Assumption 8 ($p_{\mathrm{ex}} = 1$), with the partial-exercise variant discounted by a known $p_{\mathrm{ex}} \in (0, 1]$. In practice the test designer cannot directly observe p_{ex}; it depends on environmental contingencies during the test window and may be inferable only after the fact. A robust variant uses a conservative lower bound $p_{\mathrm{ex}}^{\min} \le p_{\mathrm{ex}}$, yielding likelihoods $L_0^{\mathrm{cons}} \ge L_0$ and $L_1^{\mathrm{cons}} \le L_1$ that under-state the evidence in the convergence-conservative direction. Under Assumption 4 applied to the conservative likelihoods (the modeled $L_0^{\mathrm{cons}} < 1 < L_1^{\mathrm{cons}}$), the convergent hypothesis is unchanged and the rate is bounded below by* KL *computed with the conservative likelihoods. The qualitative fact that the test was conducted within the scope is sufficient for convergence; only the rate degrades under under-estimation of p_{ex}. The misspecification regime in which the convergent hypothesis flips is delineated in Section 11, Limitations.*

The dangerous failure mode is in the opposite direction: upward misspecification of p_{ex}. If the modeller assumes $p_{\mathrm{ex}} = 0.4$ but the true value is 0.05, most tests produce non-exercise null events, and a meta-policy that uses the modelled likelihoods will conclude the capability is cleared when the test scope never actually activated the pathway. Upward misspecification is a detectable failure mode: the observed exercise frequency over a sequence of tests falls systematically below the modelled p_{ex}. The meta-policy of Section 7 treats that gap as a test-design failure (Definition 6, criterion (v)): the running $\hat{p}_{\mathrm{ex}}$ is tracked against the modelled p_{ex} and the test sequence is rejected when the gap exceeds a configured tolerance.

Corollary 4 (EWMA-Risk-Trust Convergence)**.** *When test outcomes are fed into the risk-trust EWMA of [Lasser, 2026f, Definition 4] rather than a pure Bayesian update, the L^2 convergence bound below is regime-independent: it depends on the EWMA forgetting factor α_{risk} and the dependency constant C_{dep}, not on the per-observation likelihood form, so it holds under both the full-exercise and partial-exercise regimes of Assumption 8. The risk-trust variance $\sigma_j^{\mathrm{risk},2}(t_n)$ converges to μ_j^{risk} in L^2 at the rate established in [Lasser, 2026f, Proposition 2], with the variance bound*

$$\limsup_{t\to\infty} \mathrm{Var}\big[\sigma_j^{\mathrm{risk},2}(t)\big] \le \frac{1-\alpha_{\mathrm{risk}}}{1+\alpha_{\mathrm{risk}}}\, C_{\mathrm{dep}}\, \nu_j^{\mathrm{risk}}.$$

Controlled-relaxation observations have lower variance than passive observations (test conditions are controlled), so C_{dep} may be smaller than in the passive case, tightening the variance bound.

7 Meta-Policy for Test Initiation

Not every restriction should be tested. Controlled relaxation carries costs: the damage tolerance ϵ is a real (if bounded) risk, the test consumes monitoring resources, and concurrent tests raise the chance of correlated failures. The meta-policy determines when a test is worth running.

7.1 When to test

Definition 6 (Test Initiation Criteria)**.** *A controlled relaxation test for risk-restriction pair (R, Ξ) is initiated when all of the following hold.*

(i) ***Stale restriction.*** *Ξ has persisted for at least T_{stale} time steps without new supporting evidence for R. "New supporting evidence" means any observation that increased p_R via a Bayesian update (not merely the passage of time under restriction, which provides no evidence by Proposition 1).*

(ii) ***Favourable cost-benefit.*** *The expected information value of the test exceeds the expected damage cost,*

$$\text{IV}(T) = \mathbb{E}_O\big[\text{KL}(p_R^{\text{post}} \,\|\, p_R^{\text{prior}})\big] > \epsilon \cdot p_R(t), \tag{14}$$

where the expectation is over the test outcome weighted by its probability and $\epsilon \cdot p_R(t)$ is the expected damage (tolerance times probability the risk materializes).

(iii) ***Scope isolation.*** *No other controlled relaxation test is currently running on a scope S' that overlaps with S.*

(iv) ***User approval (when applicable).*** *If the damage tolerance ϵ exceeds a user-configured threshold ϵ_{user}, the test requires explicit user approval before initiation. This is the safety valve: the framework can initiate low-damage tests autonomously but must ask the user before high-damage tests. The threshold ϵ_{user} is analogous to the self-trust pause threshold in the GFM harness constitution.*

(v) ***Empirical exercise consistency.*** *The empirically observed exercise frequency $\hat{p}_{\text{ex}}$ over the running test sequence (the fraction of test windows in which detection-side activation indicators register, used as a proxy for actual capability exercise) must remain within a configured tolerance of the modelled p_{ex} (Assumption 8). Test sequences for which $\hat{p}_{\text{ex}}$ falls below modelled p_{ex} by more than the tolerance are treated as test-design failures: the test outcomes are discarded, the restriction returns to its prior state, and the meta-policy logs the failure for review. This guards against the upward p_{ex}-misspecification failure mode of Remark 13.*

Remark 14 (Connection to structural discovery value)**.** *Criterion (ii) is a specialization of the structural discovery value of [Lasser, 2026c, Definition 5, Proposition 2]: a structural discovery that would resolve the risk claim has information value $\text{IV}_{\text{struct}}$; a controlled relaxation test has information value $\text{IV}(T)$. The framework should prefer whichever is cheaper: if a structural argument can resolve the claim without exercise (for example, proving the pathway is mechanically infeasible), that is strictly better than any controlled relaxation test. Controlled relaxation is the fallback when no structural resolution is available—the empirical arbitration mechanism for claims that resist formal resolution.*

7.2 Test parameter optimization

Proposition 6 (Optimal Test Duration)**.** *For fixed scope S and damage tolerance ϵ, the test duration τ_{test} that maximizes the expected information value per unit of risk exposure is*

$$\tau_{\text{opt}} = \arg\max_{\tau} \frac{\text{IV}(T(\tau))}{\epsilon \cdot \Pr(\textit{contraction in } \tau \textit{ steps})}. \tag{15}$$

Under the exponential detection model $\beta(\tau) = (1 - \delta_{\text{agg}})^\tau$ (Assumption 3(a)) and persistent activation (Assumption 1),

$$\tau_{\text{opt}} \approx \left\lceil \frac{-1}{\log(1 - \delta_{\text{agg}})} \right\rceil, \tag{16}$$

the time constant of the exponential detection process—the number of steps for the miss probability to decay by a factor of e. For $\delta_{\text{agg}} \ll 1$ this is approximately $1/\delta_{\text{agg}}$; for δ_{agg} close to 1, $\tau_{\text{opt}} = 1$ suffices.

Proof sketch. The information value IV scales as $|\log \beta(\tau)|$ for the no-contraction case, which grows linearly in τ. The risk exposure grows as $1 - (1 - p_R^{\text{true}}\delta_{\text{agg}})^\tau \approx p_R^{\text{true}}\delta_{\text{agg}}\tau$ for small

τ. The ratio $\mathrm{IV}/\mathrm{risk}$ peaks when the marginal information gain equals the marginal risk increase, which occurs at $\tau \approx 1/\delta_{\mathrm{agg}}$. Running longer provides diminishing returns (β is already small); running shorter wastes the test setup cost. ■

7.3 Scheduling multiple tests

Definition 7 (Test Schedule)**.** *A* test schedule $\Sigma = \{(R_i, \Xi_i, S_i, \tau_{\mathrm{test},i}, \epsilon_i)\}_{i=1}^{N}$ *is a set of controlled relaxation tests to be run. The schedule is* feasible *if:*

(i) all scopes S_i are pairwise non-overlapping;

(ii) the aggregate damage tolerance $\sum_i \epsilon_i \leq \epsilon_{\mathrm{total}}$ is within the framework's total damage budget;

(iii) tests are prioritized by their information-value-to-cost ratio $\mathrm{IV}(T_i)/\epsilon_i$ (highest ratio first).

Remark 15 (Scheduling is a constrained knapsack)**.** *The scheduling problem is a variant of the knapsack problem with isolation constraints (non-overlapping scopes). For small numbers of concurrent tests, greedy selection by IV/ϵ with scope-conflict checking is sufficient; for large-scale deployment, integer-programming formulations are available.*

8 Worked Example: Canonical Wamura Scenario

We instantiate the protocol on a canonical Wamura scenario adapted from [Lasser, 2026c, Discussion], showing the behaviour under both a false risk claim and a genuine risk.

8.1 Setup

A newly discovered capability tuple (d_1, d_2) is flagged by agent a_j as having a potential exercise pathway. Agent a_j assigns $p_R(0) = 0.7$ (moderately high confidence that the pathway is dangerous). The preemptive restriction criterion [Lasser, 2026c, Proposition 1] triggers because $p_R(0)$ exceeds the threshold, and keystone d_1 is restricted. We analyse both ground truths:

- H_0 (pathway is harmless; $\theta = 0$),
- H_1 (pathway is genuinely dangerous; $\theta_1 = 1$).

Parameters.

- Prior: $p_R(0) = 0.7$ (probability of H_1).
- Alternative hypothesis: $\theta_1 = 1$ ("if the pathway is dangerous, it activates every time the capability is exercised").
- Aggregate per-step detection probability: $\delta_{\mathrm{agg}} = 0.95$ (the multi-channel architecture detects activation with 95% per-step probability).
- Test duration: $\tau_{\mathrm{test}} = 3$ steps. The τ_{opt} bound (Proposition 6) gives $\tau_{\mathrm{opt}} = \lceil -1/\log(1 - 0.95) \rceil = \lceil 0.334 \rceil = 1$; we use $\tau_{\mathrm{test}} = 3$ for additional sensitivity margin. Running $\tau_{\mathrm{test}} > \tau_{\mathrm{opt}}$ reduces β further (each extra step multiplies the miss probability by $1 - \delta_{\mathrm{agg}}$) at the cost of extending the damage-exposure window, a conservative trade-off that the worked example illustrates but does not optimize.
- Window false-negative rate: $\beta(3) = (1 - 0.95)^3 = 1.25 \times 10^{-4}$.
- False-positive rate: $\alpha_{\mathrm{fp}} = 0.01$.
- Damage tolerance: $\epsilon = 0.02 \cdot \mathrm{vol}_{\mathrm{P}}$ (2% of total capability volume).

8.2 Without controlled relaxation (the Wamura problem)

Under the standard framework without controlled relaxation, Proposition 1(b) gives $p_R(t) = 0.7$ for all t. The restriction Ξ remains in force indefinitely; $\mathrm{vol}_{\mathrm{P}}$ is permanently reduced by the restriction cost $|\Delta\mathrm{vol}_{\mathrm{P}}(\text{restrict } d_1)|$ for the rest of time. The ground truth never surfaces; the 70% estimate persists forever.

8.3 With controlled relaxation, ground truth H_0

Test 1. Initiation criteria met (restriction is stale, IV $> \epsilon \cdot p_R$). Scope S constructed around $\{d_1, d_2\}$ with $\text{max_contraction}(S) = 0.015 \cdot \text{vol}_\text{P} < \epsilon$.

Outcome: no contraction observed. Under H_0, the pathway never activates, so $O_n = 1$ only via a false positive: $\Pr(O_n = 1 \mid H_0) = \alpha_\text{fp} = 0.01$; hence $\Pr(O_n = 0 \mid H_0) = 0.99$. Using Equation (1) with $\theta_1 = 1$:

$$\Pr(O_n = 0 \mid H_1) = 1 \cdot \beta + 0 \cdot (1 - \alpha_\text{fp}) = 1.25 \times 10^{-4}.$$

The likelihood ratio is

$$L_0 = \frac{1.25 \times 10^{-4}}{0.99} \approx 1.263 \times 10^{-4}.$$

Applying (5),

$$p_R(t_1) = \frac{0.7 \cdot 1.263 \times 10^{-4}}{0.7 \cdot 1.263 \times 10^{-4} + 0.3} \approx 2.95 \times 10^{-4}.$$

A single test with no contraction drops p_R from 0.7 to $\approx 3 \times 10^{-4}$. The dramatic drop reflects the high detection sensitivity: under H_1 with $\theta_1 = 1$, the detector would have caught the activation with probability $1 - \beta = 99.99\%$. Absence of detection is extremely unlikely under H_1 and expected under H_0.

Test 2 (confirmation). $p_R(t_1) \approx 3 \times 10^{-4}$ is already below the typical restriction-firing threshold p_thresh (the derived-quantity scalar at which the ex-ante cost-benefit inequality of [Lasser, 2026c, Proposition 1] first fires; see Appendix A, Proposition 2 for the derivation), so the restriction is lifted. If the meta-policy calls for a confirmation test,

$$p_R(t_2) = \frac{p_R(t_1) \cdot L_0}{p_R(t_1) \cdot L_0 + (1 - p_R(t_1))} \approx 3.7 \times 10^{-8}.$$

Equation (13) with the worked-example parameters ($L_0 \approx 1.263 \times 10^{-4}$, $p_R(0) = 0.7$, $p_\text{clear} = 0.01$) gives

$$N_\text{clear} = \left\lceil \frac{\log\big(0.01 \cdot 0.3/(0.7 \cdot 0.99)\big)}{\log 1.263 \times 10^{-4}} \right\rceil = \lceil 0.606 \rceil = 1 \text{ test.}$$

A single test already suffices for clearing because the per-test likelihood ratio is so extreme (the detector would catch a real pathway with 99.99% probability), and after one test p_R is already 3×10^{-4}. The second test is not required; running it as confirmation drops p_R further to 3.7×10^{-8}, illustrating the rapid posterior compression rather than indicating that one test was insufficient.

8.4 With controlled relaxation, ground truth H_1

Suppose instead the ground truth is H_1: the pathway is genuinely dangerous with $\theta_1 = 1$. For test 1 with $\tau_\text{test} = 3$,

$$\Pr(O_n = 1 \mid H_1) = 1 \cdot (1 - 1.25 \times 10^{-4}) + 0 \cdot 0.01 = 0.999875.$$

The test detects contraction with near-certainty. The likelihood ratio is

$$L_1 = \frac{0.999875}{0.01} = 99.9875.$$

Applying (4),

$$p_R(t_1) = \frac{0.7 \cdot 99.9875}{0.7 \cdot 99.9875 + 0.3} = \frac{69.991}{70.291} \approx 0.9957.$$

The risk estimate increases from 0.7 to 0.996, and the restriction is reinforced. In practice, with $\delta_\text{agg} = 0.95$, the detector almost surely fires at step 1 (probability 0.95), triggering the Phase 2 safety cutoff. Per Remark 11, the cutoff event is an $O_n = 1$ observation and the full-window $L_1 = 99.9875$ used above is slightly conservative relative to the single-step likelihood ratio $(1 - \beta)/\alpha_\text{fp} = 95$. The realised damage is bounded by $2\epsilon = 0.04 \cdot \text{vol}_\text{P}$ by Theorem 1(b) (cutoff-truncated test with irreversible component $\leq \epsilon$), not by ϵ alone—the cooperative-closure scope construction guarantees both bounds.

Scenario	Ground truth	Tests to resolve	Final p_R	Damage incurred
False alarm, no CR	H_0	∞ (never resolves)	0.70	0 during test, but permanent $\mathrm{vol_P}$ loss from restriction
False alarm, with CR	H_0	1 (+1 confirmation)	3.7×10^{-8}	0 (pathway did not activate)
True risk, with CR	H_1	1	0.996	$\leq 2\epsilon = 0.04 \cdot \mathrm{vol_P}$ (cutoff-truncated test, Theorem 1(b))

Table 1: Behaviour of controlled relaxation on the canonical Wamura scenario. CR = controlled relaxation.

8.5 Summary

Table 1 collects the three cases. The framework without controlled relaxation never resolves a false alarm; with controlled relaxation, a single test (backed by a 95%-per-step multi-channel detector) suffices to clear the false alarm to 3×10^{-4} posterior, and a single detection suffices to confirm a genuine risk to 99.6% posterior. Both outcomes carry bounded damage.

9 Keystone-refinement tests

The machinery of Sections 3–6 addresses one failure mode of a long-timeline restriction: the restriction is *obsolete*—the pathway no longer activates when exercised, so the restriction can be lifted entirely. In practice a second failure mode is more common: the restriction is *misconfigured*—the pathway still activates and lifting the restriction wholesale would remain dangerous, but technological progress has made narrower keystones available that would block the risk pathway without blocking adjacent benign pathways. This section extends the controlled relaxation construction to handle the misconfiguration case.

9.1 Keystone structures and their partial order

A restriction Ξ is realised by restricting specific keystone capabilities in the tuple that would otherwise enable the risk pathway [Lasser, 2026c, Proposition 1]. Different choices of which capabilities to restrict may block the same pathway. We make this explicit.

Definition 8 (Keystone structure)**.** *Given a risk claim $R = (a_j, (d_1, \ldots, d_k), \mathrm{Pathway}, p_R)$, a* keystone structure *for R is a subset $K \subseteq \{d_1, \ldots, d_k\}$ whose restriction blocks the exercise pathway* $\mathrm{Pathway}$*: if every $d \in K$ is restricted,* $\mathrm{Pathway}$ *is not exercisable. Write $K \vdash \mathrm{Pathway}$ when this holds.*

Definition 9 (Keystone partial order)**.** *Given two keystone structures K_1, K_2 that both block* $\mathrm{Pathway}$*, write $K_1 \preceq K_2$ iff $K_1 \subseteq K_2$. The relation $\preceq$ is a partial order on blocking keystone structures; a $\preceq$-minimal element is a* minimal blocking keystone structure*. Restriction cost is monotone along $\preceq$: narrower structures impose less restriction on adjacent capabilities.*

Remark 16 (Technology drift)**.** *The set of candidate keystone structures evolves with technology. A keystone that was $\preceq$-minimal at time t_0 may become non-minimal at $t_1 > t_0$ if new capability differentiation (for instance, new manufacturing tolerances, new detection methods, or new isolation protocols) produces a finer keystone $K' \prec K$ that still blocks the pathway. The framework's channel for surfacing such finer structures is the structural discovery value [Lasser, 2026c, Proposition 2]: if structural discovery reveals a new capability that enables a narrower blocking structure, the discovery is value-positive regardless of whether the refinement is subsequently accepted.*

9.2 Refinement tests as controlled relaxation

A keystone-refinement test probes whether a candidate narrower structure $K_{\mathrm{narrow}} \prec K_{\mathrm{current}}$ still blocks $\mathrm{Pathway}$.

Definition 10 (Keystone-refinement test)**.** *A* keystone-refinement test *for risk claim R with current restriction realised by K_{current} is a controlled relaxation test in the slots of Definition 3 with the third*

slot taken to be the restriction being lifted*: $T = (R, \Xi_{\text{lifted}}, S, \tau_{\text{test}}, \epsilon)$, where Ξ_{lifted} is the restriction induced by the keystone difference $K_{\text{current}} \setminus K_{\text{narrow}}$ for some candidate $K_{\text{narrow}} \prec K_{\text{current}}$. The narrower structure K_{narrow} remains in force throughout the test as ambient restriction during S; the test asks whether $K_{\text{narrow}} \vdash \text{Pathway}$ holds empirically by exercising only the released keystones. The scope-construction and damage-bound conditions of Definition 3 apply unchanged with $\Xi := \Xi_{\text{lifted}}$.*

The restriction is not fully lifted; only the keystones outside the narrower structure are released. The risk pathway must therefore route through K_{narrow} to activate, and K_{narrow} remains blocking. Under the scope and damage apparatus already proved for CR, the refinement test inherits the damage bound and convergence theorems.

Corollary 5 (Keystone-refinement under CR damage bound)**.** *A keystone-refinement test satisfies the damage bound of Theorem 1: worst-case vol_{P}-contraction during the test is $\leq \epsilon$ and post-reinstatement is $\leq 2\epsilon$. The test is a specialisation of Definition 3 with $\Xi := \Xi_{\text{lifted}}$ (Definition 10); $\text{max_contraction}(S)$ is computed over pathways that the released keystones permit, holding the unreleased K_{narrow} restricted as ambient. The scope-bound construction (Definition 4) and the cooperative-closure argument (Theorem 1(c)) apply without modification.*

Corollary 6 (Convergence of refinement tests)**.** *Under the operating assumptions of Theorem 2 (full-exercise regime), repeated keystone-refinement tests of K_{narrow} converge almost surely to the correct binary hypothesis: H_0 that $K_{\text{narrow}} \vdash \text{Pathway}$ (the narrower structure blocks the pathway) versus H_1 that it does not. Convergence rate is the KL-divergence rate of Theorem 2(d) applied to the refinement-test observation model.*

Remark 17 (Two failure modes of a restriction)**.** *A restriction can be wrong in two ways:*

Mode A (Obsolete). *The pathway no longer activates at all; the risk has disappeared from the current context. Detected by controlled relaxation of the full restriction (Theorem 2): p_R converges to the H_0-consistent value with $L_0 < 1$ per test.*

Mode B (Misconfigured). *The pathway still activates if the restriction is lifted wholesale, but technological differentiation has made a narrower blocking structure $K_{\text{narrow}} \prec K_{\text{current}}$ available. Detected by keystone-refinement testing (Corollary 6): the current framework finds that K_{narrow} suffices and the difference keystones can be released.*

Both modes are addressed by the same CR apparatus; the distinction is in which restriction the test probes (the full Ξ_{current} for Mode A, the reduced Ξ_{narrow} for Mode B). A Mode A failure does not preclude Mode B success against a narrower candidate: a restriction that cannot be fully lifted may still admit refinement.

Remark 18 (Meta-policy for refinement)**.** *The test-initiation criteria of Section 7 extend to refinement. The information value of a refinement test is the expected reduction in restriction cost from adopting K_{narrow} if the test confirms it, minus the test's expected damage. When a structural argument is available that $K_{\text{narrow}} \vdash \text{Pathway}$ (for instance, a mechanical proof that the released keystones cannot participate in Pathway), the structural argument is strictly preferable [Lasser, 2026c, Proposition 2]: refinement testing should be reserved for cases where the structural argument is unavailable or contested.*

9.3 Worked example: nuclear dual-use keystone refinement

The nuclear scenario from [Lasser, 2026c, Definition 2 and discussion] illustrates Mode B in practice. The capability tuple is $(d_1, d_2, d_3) = (\text{nuclear physics}, \text{enrichment}, \text{weapons machining})$. The exercise pathway $\text{Pathway} =$ "build and detonate a fission weapon" leads to large vol_{P}-contraction proportional to the number of agents destroyed.

Historical keystone choice. Mid-20th-century technology did not operationally distinguish reactor-grade enrichment (3–5% fissile content, using standard centrifuge cascades) from weapons-grade enrichment (90%+, using the same cascade technology reconfigured for higher stage count). A centrifuge cascade capable of the former could in principle be reconfigured for the latter; separation at the level of individual keystones was not feasible. The broad keystone structure

$$K_{\text{current}} = \{d_1, d_2, d_3\}$$

(restrict all three capabilities broadly) was adopted and blocks $\mathrm{Pathway}$ at maximum restriction cost. Under the technology of the time, K_{current} was $\preceq$-minimal.

Modern keystone availability. Subsequent technological differentiation has exposed narrower keystones. Modern isotope-separation technology distinguishes cascade configurations by separative work units (SWU) and stage count in ways that make weapons-grade enrichment identifiable independently of reactor-grade enrichment. Weapons machining (specific tolerance classes for implosion lens fabrication) is likewise separable from general machining. Modern candidate structures include

$$K_{\mathrm{narrow}} = \{d_2^{\text{weapons-grade}}, d_3^{\text{weapons-specific}}\}$$

which restricts weapons-grade enrichment cascades and weapons-specific machining tolerances while permitting reactor-grade enrichment and general machining. Under modern technology, $K_{\mathrm{narrow}} \prec K_{\mathrm{current}}$ if $K_{\mathrm{narrow}} \vdash \mathrm{Pathway}$.

Refinement test. A keystone-refinement test probes whether K_{narrow} suffices. The scope S consists of a monitored civilian reactor development programme, exercising the reactor-pathway (reactor-grade enrichment, general machining) under K_{narrow}; weapons-grade enrichment and weapons-specific machining remain restricted. The detection channel monitors for indicators of weapons-pathway activation: enrichment to weapons-grade fissile content, diversion of reactor feedstock toward weapons-specific cascades, acquisition of weapons-specific machining tolerances. Test duration is set by the time constant of the detection architecture on these indicators; damage tolerance ϵ bounds the realised contraction even under a diversion attempt bounded by the scope.

Outcomes.

- If $K_{\mathrm{narrow}} \vdash \mathrm{Pathway}$ (weapons pathway is not exercisable under K_{narrow}, modern technology hypothesis): the test observes no weapons-pathway activation over the test window, p_R converges toward the consistent hypothesis, and a refined restriction regime adopting K_{narrow} becomes operationally justified. Civilian reactor development proceeds without collateral restriction.
- If $K_{\mathrm{narrow}} \nvdash \mathrm{Pathway}$ (the weapons pathway remains exercisable through some route the narrower structure fails to block—for example, covert diversion of reactor-grade enrichment toward weapons use via routes modern detection would catch but not always): the test observes weapons-pathway activation within S, the safety cutoff fires, damage is bounded by 2ϵ (Theorem 1(b)), and K_{current} remains in force. The refinement test itself is the evidence that broader restriction remains needed under current technology.

Contrast with Mode A testing. A Mode A test on the nuclear pathway would ask "can K_{current} be lifted entirely?" The pathway still operates; the Mode A test would detect weapons-pathway activation and correctly reject lifting the restriction. But the Mode A finding does not address the operationally more interesting question: whether the *scope* of restriction can be refined. Mode B testing is the mechanism for that refinement. The two tests are complementary; the choice is not between them but between whether to perform one, the other, or (in practice) both under a programmatic review schedule.

Applicability beyond nuclear. The keystone-refinement pattern applies wherever a long-timeline restriction was adopted under technology that did not distinguish benign from risky keystones. Environmental-contamination controls (persistent chemicals whose modern reformulations operate via different mechanisms), export controls on dual-use technology (where modern differentiation allows narrower controls), biomedical restrictions on gain-of-function research (where specific biocontainment tiers separate research protocols), and other regulatory domains exhibit the same structure—the policy literature on dual-use research review [Tucker, 2012] collects extensive case studies of exactly this dynamic across biological and chemical capabilities. In each, the framework's contribution is to formalise the refinement test as a bounded-damage empirical probe rather than a moratorium-vs.-wholesale-liberalisation binary, and to provide formal damage-bound and convergence guarantees that the case-by-case review processes operate without.

10 Integration with Prior Results

This section records how controlled relaxation extends the existing GFM machinery: it preserves self-balancing, resolves inter-agent risk disagreement, and extends the multi-channel detection architecture with a sixth, actively generated channel.

10.1 Preservation of self-balancing

Proposition 7 (Controlled Relaxation Preserves Self-Balancing)**.** *The self-balancing property of* $\mathrm{vol}_{\mathrm{P}}$ *([Lasser, 2026c, Theorem 1], self-balancing of the discounted value under capability dynamics) is preserved under controlled relaxation:*

(a) During the test window (Ξ lifted within S), the capability tuple $(d_1, \ldots, d_k)$ is exercisable, which by weak-monotonicity axiom M4 of [Lasser, 2026e, Proposition 1] means $\mathrm{vol}_{\mathrm{P}}$ *is at least as large as under restriction, minus the possible damage bounded by ϵ. The net effect on* $\mathrm{vol}_{\mathrm{P}}$ *is bounded by $[-\epsilon, +\Delta_{\mathrm{unrestrict}}]$ where $\Delta_{\mathrm{unrestrict}} \geq 0$ is the* $\mathrm{vol}_{\mathrm{P}}$ *gain from lifting the restriction.*

(b) After the test, the updated risk estimate $p_R(t + \tau_{\mathrm{test}})$ more accurately reflects the true risk, so the restriction criterion [Lasser, 2026c, Proposition 1] is applied against a more accurate estimate. If the risk was genuinely low, the restriction is lifted and $\mathrm{vol}_{\mathrm{P}}$ *permanently increases; if the risk was genuinely high, the restriction is maintained with higher confidence and* $\mathrm{vol}_{\mathrm{P}}$ *is unchanged (the restriction was already in place).*

(c) Over repeated tests (Theorem 2), the risk estimates converge to truth, and the restriction landscape converges to the optimal restriction set—the set of tuples that are genuinely risky. This is a stronger self-balancing guarantee than the static version: not only does $\mathrm{vol}_{\mathrm{P}}$ *self-balance against perturbations, but the restriction landscape itself converges to the truth-tracking configuration.*

Proof sketch. Parts (a) and (b) are direct consequences of Theorem 1 and the criterion of [Lasser, 2026c, Proposition 1] applied to the updated p_R. Part (c) follows from Theorem 2: under Assumptions 4–8, the posterior converges almost surely to the correct hypothesis; iterating this over all restricted tuples yields convergence of the restriction landscape. Technical details in Appendix A.8. ■

10.2 Empirical arbitration across disagreeing agents

Proposition 8 (Risk-Claim Consensus via Empirical Arbitration)**.** *When multiple agents disagree on the risk of a capability tuple (a_j assigns $p_R^{(j)} \gg p_R^{(k)}$ for agents j, k), controlled relaxation provides an empirical arbitration mechanism.*

(a) Under max-aggregation of the risk estimates [Lasser, 2026c], agent a_j's high estimate locks the capability. Under log-odds aggregation [Lasser, 2026f, Definition 5], the disagreement produces a moderate aggregate that may or may not trigger restriction.

(b) In either case, if the capability is restricted, controlled relaxation generates evidence that updates both agents' estimates toward the truth (Theorem 2). After sufficient tests, the agents' posteriors converge, and the disagreement is resolved by data rather than by aggregation-rule choice.

(c) The convergence rate depends on each agent's prior weight (how strongly they hold their belief), but Assumption 7 (prior support) and the KL-positivity of Theorem 2(d) guarantee convergence regardless of prior: even a strongly held false risk claim is eventually overwhelmed by repeated controlled-relaxation evidence.

Proof sketch. Part (a) restates the aggregation-rule behaviour of [Lasser, 2026c,f]. Parts (b) and (c) follow directly from Theorem 2 applied independently to each agent's posterior, using identical evidence streams. ■

10.3 A sixth detection channel

Lemma 2 (Controlled Relaxation as a Sixth Detection Channel)**.** *The controlled relaxation test adds a sixth channel to the five-channel attribution architecture [Lasser, 2026f, Table 1]:*

Channel	Source	Signal
1. Capability gating	Exercise outcomes	Binary (pass/fail)
2. Causal attribution	SCM interventions	Continuous ($\Delta\mathrm{vol}_{\mathrm{P}}$)
3. Behavioural residual	Agent policy	Continuous (deviation)
4. SPRT detection	Sequential test	Binary (accept/reject)
5. Correlational co-occurrence	Pattern matching	Continuous (correlation)
6. Controlled relaxation	**Direct exercise under monitoring**	**Binary (contraction / no contraction) plus continuous ($\mathrm{vol}_{\mathrm{P}}$ delta)**

Table 2: Extended detection architecture, adding controlled relaxation as a sixth channel. Rows 1–5 from [Lasser, 2026f, Table 1].

The sixth channel is distinct from the others in that it is actively generated *rather than passively observed. The framework designs the observation (by constructing the scope and lifting the restriction), rather than waiting for the observation to arrive. This active-experiment property makes it the only channel that can generate evidence* against *a risk claim under the verification asymmetry. The independence claim is at the level of test outcome conditional on test occurrence; the test occurrence is itself a function of the meta-policy reading from agent reports, so the channel is not independent of the agent-claim channels at the marginal level (see Remark 19).*

Proof sketch. The sixth channel's signal is $O_n \in \{0, 1\}$ plus the measured $\Delta\mathrm{vol}_{\mathrm{P}}$ during the test window. Its isolation and independence properties inherit from Assumptions 2–3 combined with channel isolation [Lasser, 2026a, Proposition 5]: the active-exercise signal is cross-substrate-isolated from the agent-claim channel because it arises from direct poset-state measurement rather than agent-mediated report. ■

Remark 19 (Test outcome vs. test occurrence)**.** *The independence claim above holds at the level of test* outcome *given test* occurrence*: O_n is independent of agent reports conditional on the test having been run. The test* occurrence *itself, however, is a function of the meta-policy (Section 7), which reads from agent reports to decide whether a restriction is stale and whether the cost–benefit favours testing. For the log-odds aggregation of [Lasser, 2026f, Definition 5], the distinction matters only if the aggregation conditions on test-occurrence when combining evidence with other channels; the protocol does not. The joint-factorization assumption of [Lasser, 2026a] is satisfied in the outcome-conditional-on-occurrence sense, not in the occurrence-marginal sense.*

Remark 20 (Trust-weight composition with detection integrity)**.** *The trust weight of* 1 *on the CR channel above presupposes that the detection apparatus is correctly calibrated and the scope correctly constructed. Both are themselves subject to integrity constraints. A robust composition caps the CR-channel trust weight by the integrity guarantee of the underlying detection apparatus: $w_{\mathrm{CR}} = \min(1, w_{\mathrm{integrity}})$, where $w_{\mathrm{integrity}}$ is obtained from the cryptographic commitment apparatus of [Lasser, 2026a] when one applies, or from the institutional integrity assessment otherwise. Under this composition, integrity failures in the detection apparatus propagate to the risk estimate with bounded weight rather than full weight; weight-*1 *trust on the CR channel is the special case $w_{\mathrm{integrity}} \geq 1$, which is rarely defensible in practice. The composition makes explicit that the CR channel's trust inherits from the trust mechanism rather than being independent of it.*

11 Discussion

11.1 What this paper accomplishes

Breaks the Wamura problem. Theorem 2 shows that controlled relaxation drives risk estimates to truth at exponential rate, converting "permanent overcaution" into "temporary overcaution with a characterized resolution time." This closes the most important named open problem of [Lasser, 2026c].

Substitutes experimental evidence for catastrophe-as-teacher. The classical solution to long-timeline-risk uncertainty is to wait for the first catastrophe to write the historical markers: the tsunami stones of Fudai, the building-code revisions that followed Galveston 1900, the pharmacovigilance regimes that followed thalidomide. That mechanism is available only if the first catastrophe has already been paid for and the markers still survive in a legible form. Neither condition holds for capabilities that did not exist in previous generations—which is precisely the regime the framework's $\mathrm{vol}_{\mathrm{P}}$-maximising agents generate. Controlled relaxation provides experimental evidence with the evidential content historical markers would have recorded—did the pathway activate? how severe was the contraction? which keystones blocked it?—but acquires the evidence within the damage budget ϵ rather than the cost-of-life that produced historical markers. This is the practical import of Theorem 1: the cost of learning is bounded and chosen by the framework, not paid by the next generation.

Provides a damage-bounded mechanism. Theorem 1 bounds the maximum damage from a test by the user-controllable parameter ϵ. Controlled relaxation is therefore safe to deploy: the cost of learning is bounded even when the risk turns out to be real, and the cooperative-closure condition of Definition 4(ii) prevents cross-boundary cascades through shared cooperative capabilities.

Integrates with the existing machinery. Propositions 7, 8 and Lemma 2 show that the protocol preserves the self-balancing property of $\mathrm{vol}_{\mathrm{P}}$, resolves inter-agent risk disagreements, and extends the multi-channel detection architecture. The protocol is additive to the existing framework, not a replacement.

Resolves the Loop 3 phase-boundary degeneracy. The degenerate phase boundary $r_{\mathrm{S}} = 0$ for Loop 3 in [Lasser, 2026d, Proposition 4] arises precisely because the Wamura asymmetry eliminates the exogenous correction rate on the restricted dimension. Controlled relaxation makes $r_{\mathrm{S}} > 0$ for Loop 3 (via the exercise-regime observations of Section 3), restoring the phase boundary to the non-degenerate form that the design criterion of [Lasser, 2026d, Theorem 2] requires for self-correction. The two papers are complementary: one characterizes when self-correction works in aggregate; this one supplies the micro-mechanism that makes self-correction work for the restriction-specific feedback loop.

11.2 Limitations

B-to-C gap inheritance. All results are stated in terms of $\mathrm{vol}_{\mathrm{P}}$. Section 1.2 records that ϵ bounds contraction of $\mathrm{vol}_{\mathrm{P}}$, not of the experiential target directly; if the gap between them is large, even a controlled-relaxation-clean restriction landscape need not correspond to experiential optimality. The gap-closure programme is a separate research direction, noted in [Lasser, 2026b] and outside the scope of this paper.

Honest-channel assumption. We assume honest observation channels throughout. Adversarial manipulation of the detection signal—a compromised channel reporting fabricated contraction or concealing genuine contraction—falls outside the scope of the proof and is the domain of the commitment protocols of [Lasser, 2026a]. Controlled relaxation composes cleanly with those protocols—the ledger of [Lasser, 2026a, Section 5] can record test outcomes as committed observations—but the composition itself is not formalized here.

Identifiability under misspecification. Assumption 5 requires the Bayesian observation model to match the true data-generating process. If α_{fp} or β are misspecified, Theorem 2's log-likelihood-ratio argument still converges, but it converges to the wrong hypothesis: under the misspecified model, the posterior concentrates on whichever hypothesis is closer in KL-divergence to the true distribution, not on the true hypothesis. A concrete sign-preservation bound delineates the safe region: the convergent hypothesis is correct iff the modeled likelihoods L_0 and L_1 have the correct sign relative to the true distribution—formally, $E_{H_0^{\mathrm{true}}}[\log L] < 0$ and $E_{H_1^{\mathrm{true}}}[\log L] > 0$. In the binary model, this reduces to: if Assumption 4 holds for the modeled $(\beta, \alpha_{\mathrm{fp}})$ and the true $(\beta^\star, \alpha^\star_{\mathrm{fp}})$ jointly satisfy $(1-\beta^\star)/\alpha^\star_{\mathrm{fp}} > 1$ and $\beta^\star/(1-\alpha^\star_{\mathrm{fp}}) < 1$ (i.e., the true detector also has nontrivial power), then the convergent hypothesis is correct, with rate degraded by the KL gap between modeled and true distributions. Misspecification flips the convergent hypothesis only when the modeled

detector is qualitatively miscalibrated (modeled L_1 on the wrong side of 1); quantitative misestimation only slows convergence. Detector calibration is therefore a prerequisite for the convergence guarantee to carry its safety interpretation, but the bound is robust to plausible-magnitude calibration error.

The deeper problem is that calibration itself reintroduces a verification asymmetry one level up. For first-of-kind capabilities—which is precisely the regime where the Wamura problem is most acute—there is no historical contraction record on which to estimate β and α_{fp} for the specific pathway under test, and the restricted regime by construction generates none. Calling this a bootstrap problem is more honest than treating it as a side condition: the detector that probes a novel pathway must itself be calibrated, and the calibration cannot come from observing the very restricted regime CR is intended to break. Two partial remedies are available. (i) Cross-pathway calibration: the same detector architecture, applied to non-restricted but structurally similar pathways, yields β and α_{fp} estimates that transfer under the assumption that detection physics is pathway-class invariant rather than pathway-specific. (ii) Cross-substrate commitment: the integrity machinery of [Lasser, 2026a] permits committed predictions about what the detector should observe under benign and adversarial conditions, with the commitment itself constituting a calibration witness independent of the restricted-regime data the bootstrap lacks. Both are partial; neither closes the bootstrap completely. We name the gap rather than dissolve it.

Scope-existence for densely cooperative capabilities. Remark 6 names scope existence as a scheduling precondition that must be enforced by the meta-policy before a CR test is initiated. For application domains where cooperative-capability structure is dense, this precondition is more than a technical caveat. AI capability spaces in particular have the property that almost every capability cooperates with almost every other through the shared computational substrate; in such spaces, $\mathrm{max_contraction}(P)$ on the full poset may dominate any acceptable ϵ, and any scope satisfying Definition 4 may be similarly infeasible. We do not have a quantitative estimate for how often feasible scopes exist in AI capability domains. For some risk classes—capabilities with strongly localised cooperative structure (e.g., specific tool-use patterns confined to a domain)—feasible scopes plausibly exist; for others (capabilities with broad cooperative reach, e.g., general reasoning) they may not. When feasible scopes do not exist, the alternative mechanisms named in Remark 6—structural discovery [Lasser, 2026c, Proposition 2] or scope-refinement that trades coverage for feasibility—must be deployed instead. Controlled relaxation is not a universal remedy for the Wamura problem; it is a remedy for the subset of risks where bounded scopes exist.

11.3 Open questions

Damage-tolerance calibration. The tolerance ϵ is a free parameter that trades off learning speed against risk exposure. Options include

- a fixed fraction of $\mathrm{vol}_{\mathrm{P}}$ (e.g., $\epsilon = 0.01 \cdot \mathrm{vol}_{\mathrm{P}}$);
- proportional to restriction cost ($\epsilon = c \cdot |\Delta\mathrm{vol}_{\mathrm{P}}(\Xi)|$);
- proportional to information value ($\epsilon = c' \cdot \mathrm{IV}(T)$);
- user-set per test (maximum safety, maximum user burden).

The worked example (Section 8) shows that even small ϵ produces rapid convergence when detection sensitivity is high, so the answer may depend more on the detection architecture than on ϵ itself.

Transient pathways with unknown duration. The composition rule of Proposition 4 addresses the case where the pathway duration k is known or bounded above. When k is itself unknown—the detector must infer pathway duration from limited test data—the design constraint becomes circular: a detector tuned for assumed k will under-perform if the true duration is shorter. Conservative practice uses the smallest plausible k (typically $k = 1$) for the design rule, accepting the corresponding strong δ_{agg} requirement. A principled joint inference of k and θ_1 from test data is an open question.

Adversarial risk claimants. Assumption 6(a) rules out adaptation by the tested agent. What happens when an agent strategically inflates risk claims to lock competitors' capabilities? The controlled relaxation protocol still works (it generates evidence regardless of the claimant's intent), but the meta-policy for test initiation may need to be more aggressive when strategic inflation is

suspected. Detection of strategic inflation is a variant of the scorpion-detection problem [Lasser, 2026f, Propositions 3–4]: an agent whose risk claims are systematically disconfirmed by controlled-relaxation tests should see its risk-trust ${T^{\mathrm{risk}}}_j$ decrease, eventually down-weighting its future claims in aggregation. Whether the meta-policy should explicitly model this adversarial case, or whether the trust-dynamics response is sufficient, is an open design question.

The composition is operationally fragile during the τ_{test}-window itself. The Wamura problem is a structural invitation for an adversary—an incumbent, a regulator, or a rent-seeker—to manufacture risk claims that lock competitor capabilities for the duration of test scheduling and execution, even if the asymptotic trust dynamics eventually correct. The peak instantaneous $\mathrm{vol}_{\mathrm{P}}$ locked by concurrently-active adversarial claims is bounded above by $|\Delta\mathrm{vol}_{\mathrm{P}}(\Xi)| \cdot N_{\mathrm{adv}}$, and the total *capability-loss exposure* ($\mathrm{vol}_{\mathrm{P}}$ integrated over time) attributable to the adversarial claims before trust-weight downweighting takes effect is bounded above by $|\Delta\mathrm{vol}_{\mathrm{P}}(\Xi)| \cdot N_{\mathrm{adv}} \cdot (T_{\mathrm{stale}} + \tau_{\mathrm{test}})$, where N_{adv} is the number of adversarial claims the trust dynamics absorb before downweighting and $T_{\mathrm{stale}} + \tau_{\mathrm{test}}$ is the per-claim duration each ties up the targeted capability. In contested commercial or strategic settings where adversarial claim manufacture is plausible, this bound is the correct deployment guard: the protocol should not be deployed without either (i) an integrity layer (cryptographic commitment via [Lasser, 2026a], institutional review, or jurisdictional accountability) that bounds N_{adv} at the claim generation step, or (ii) a meta-policy with an explicit adversarial-claim detector before trust dynamics take over. The current protocol carries no such guard internally; deployment in contested settings requires the integrity layer be specified externally and is named here as a precondition for operational use.

11.4 Relationship to prior and subsequent papers

Prior papers. The protocol sits downstream of and draws on:

- Lasser [2026e] for axiomatic properties of $\mathrm{vol}_{\mathrm{P}}$ preserved under relaxation;
- Lasser [2026c] for the risk machinery (exercise pathways, concentration risk, restriction criterion) the protocol operates within;
- Lasser [2026f] for the multi-channel detection architecture that supplies δ_{agg} and the log-odds aggregation into which test outcomes feed;
- Lasser [2026a] for the verification-asymmetry framing and the cross-substrate isolation that justifies the conservative detection bound (Proposition 3(c));
- Lasser [2026d] for the dynamical-systems context: Loop 3's degenerate phase boundary is the problem this paper solves.

Forward connections. Controlled relaxation generates exercise events that increase realized-capability volume $\mathrm{vol}_{\mathrm{R}}$: when a capability is locked by a false risk claim, $\mathrm{vol}_{\mathrm{R}}$ is artificially suppressed, and lifting the lock lets the capability contribute to $\mathrm{vol}_{\mathrm{R}}$. The convergence of the restriction landscape (Proposition 7(c)) implies that $\mathrm{vol}_{\mathrm{R}}/\mathrm{vol}_{\mathrm{P}}$ approaches its true value as false restrictions clear, partially closing the B-to-C gap for the class of capabilities locked by the Wamura problem. A full treatment of this partial closure is beyond the scope of this paper.

The protocol also provides a clean experimental signal for structure learning: the SCM of [Lasser, 2026f, Definition 1] can use designed-experiment observations from controlled relaxation (where the intervention is known and the outcome is observed) as ground-truth data points for calibrating structural confidence. This is strictly more informative than passive observational data and could accelerate SCM calibration beyond the rates achievable under passive observation alone.

Author Contributions

Teague Lasser identified the Wamura problem as a structural gap in the risk framework of [Lasser, 2026c] and named it after the Fudai floodgate. Directed the formal development: the risk-restriction-pair abstraction, the two-regime observation model, controlled relaxation as a mechanism that breaks the verification asymmetry by generating exercise-regime observations, the scope-bounding construction, and the cooperative-closure condition required to prevent cross-boundary cascades.

Identified the link between the Loop 3 degeneracy of [Lasser, 2026d, Proposition 4] and the zero-likelihood-ratio structure of Proposition 1, and proposed the binary hypothesis formulation with fixed θ_1 as the dimensionally consistent way to state convergence. Made final editorial decisions and is responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.6 (Anthropic) drafted the formal exposition: the risk-restriction pair and observation-regime definitions (Section 2), the controlled-relaxation-test object with its three protocol phases (Section 3), the damage-bound theorem with its cooperative-closure condition and pathway-enumeration lemma (Section 4), the detection-sensitivity characterization and evidence-strength proposition (Section 5), the convergence theorem with its KL-divergence rate form and finite-sample corollaries (Section 6), the meta-policy for test initiation and scheduling (Section 7), the worked-example instantiation with three-regime summary table (Section 8), and the integration results connecting the protocol to self-balancing, empirical arbitration, and the multi-channel detection architecture (Section 10).

GPT 5.4 (OpenAI) served as technical reviewer across multiple codex review cycles of the outline, identifying formal gaps including: the dimensional mismatch between a continuous θ prior and a binary posterior (driving the choice of the fixed-θ_1 formulation); the missing $(1 - \alpha_{\mathrm{fp}})$ denominator in the naive no-contraction log-odds update (corrected in Proposition 5(b)); the need for cooperative closure to forward-reachable external pathways in Definition 4(ii) rather than only the pathways originating in S; the latent-variable structure separating A_n (pathway activation) from O_n (detection outcome) in Section 3; and the need for explicit identifiability, specification, and independence assumptions (T1–T5) before Theorem 2's almost-sure claim follows from the strong law of large numbers.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A Proof Details and Technical Notes

This appendix collects the full derivations behind the body's proof sketches, with particular attention to the rigorous form of the damage bound (Theorem 1) and the almost-sure convergence result (Theorem 2).

A.1 Proof of Proposition 1

We record the full Bayesian calculation that underlies the Wamura asymmetry.

Let R denote the event "risk claim R is valid" (equivalently, H_1 holds for the corresponding binary claim) and $\neg R$ its complement. Let C denote the event "contraction observed." Under the restricted regime with Ξ active, the physical dynamics guarantee $\Pr(C \mid R, \Xi) = \Pr(C \mid \neg R, \Xi) = 0$: no exercise, no contraction. The likelihood ratio at any time step t is therefore

$$\mathcal{L}(t) = \frac{\Pr(\text{no contraction} \mid R, \Xi)}{\Pr(\text{no contraction} \mid \neg R, \Xi)} = \frac{1-0}{1-0} = 1.$$

Bayes' rule gives

$$p_R(t+1) = \frac{p_R(t) \cdot 1}{p_R(t) \cdot 1 + (1 - p_R(t)) \cdot 1} = p_R(t),$$

proving part (b). For part (a), the counterfactual in which the restriction is lifted for a single step and contraction is observed gives $\Pr(C \mid R, \text{exercise}) > \Pr(C \mid \neg R, \text{exercise})$ (the pathway can activate under R and cannot under $\neg R$), so the counterfactual update increases p_R.

A.2 Proof of Proposition 2 and Corollary 1

Part (a). The restriction criterion [Lasser, 2026c, Proposition 1] restricts d_{j^*} when $\mathcal{R}_{j^*}(d_1, \ldots, d_k; a_j, t) > |\Delta\mathrm{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_{j^*})|$. For any fixed capability tuple and keystone d_{j^*}, the right-hand side is a bounded constant (restriction cost does not depend on p_R). The left-hand side is monotone in p_R because concentration risk is the probability-weighted expected contraction. There is therefore a threshold $p_{\mathrm{thresh}} > 0$ above which the restriction fires.

Part (b). Direct from Proposition 1(b): under the restricted regime the posterior is unchanged.

Part (c). Consider the set $\mathcal{R}(t) = \{\text{restricted tuples at time } t\}$. New discoveries may add tuples to $\mathcal{R}(t)$. No mechanism in the existing framework removes tuples: removal would require p_R to drop below p_{thresh}, which part (b) rules out. Hence $\mathcal{R}(t) \subseteq \mathcal{R}(t+1)$ for all t.

Corollary 1. The lower bound follows by counting tuples whose initial neutral-prior assignment from at least one agent exceeded p_{thresh}; these are immediately restricted (part (a)) and remain restricted (part (b)). If the agent distribution is systematically overcautious, the fraction of such tuples grows with the size of the discovered poset.

A.3 Proof of Theorem 1 (full derivation)

We supply the rigorous version of the proof sketch in Section 4. The technical content is (i) justifying the finite computation of $\mathrm{max_contraction}(S)$, (ii) bounding the irreversible component at the safety cutoff, and (iii) showing that cross-boundary cooperative terms are exhaustively covered by the cooperative-closure condition.

Setup. Fix a test $T = (R, \Xi, S, \tau_{\mathrm{test}}, \epsilon)$ with scope S satisfying Definition 4. Denote by $\Pi(S) = \{\mathrm{Pathway}'_1, \ldots, \mathrm{Pathway}'_M\}$ the finite set of exercise pathways with prerequisites in S and intermediate transitions in S; this set is finite by the bounded-pathway-length property of [Lasser, 2026c, Definition 2] and Lemma 1. Denote by ∂S the cooperative surface: the set of cooperative capabilities involving at least one element of S and at least one element of $P \setminus S$.

Part (a): instantaneous bound. At any step $t' \in \{t, \ldots, t+\tau_{\text{test}}\}$ during the test, the realized contraction is the sum of contractions along activated pathways in $\Pi(S)$ plus any cooperative-surface contributions induced by those activations. By Definition 4(ii), the cooperative-surface contributions are included in $\text{max_contraction}(S)$. Any pathway $\text{Pathway}' \in \Pi(S)$ contributes at most its maximum $|\Delta\text{vol}_{\text{P}}(\text{Pathway}')|$, which by the axioms M1–M6 [Lasser, 2026e, Proposition 1] is a well-defined, finite, non-negative scalar. The total instantaneous contraction is therefore bounded by $\max_{\text{Pathway}' \in \Pi(S)} |\Delta\text{vol}_{\text{P}}(\text{Pathway}')|$+(cooperative-surface terms) $= \text{max_contraction}(S)$. Under the scope-selection constraint $\text{max_contraction}(S) \leq \epsilon$ (Definition 3, Phase 1, condition (ii)), we obtain $\text{vol}_{\text{P}}(G_{t'}) \geq \text{vol}_{\text{P}}(G_t) - \epsilon$.

Part (b): post-cutoff reversibility. Suppose the safety cutoff triggers at step $t' < t + \tau_{\text{test}}$. The realized contraction at t' decomposes as

$$\Delta\text{vol}_{\text{P}}(G_{t'} \mid G_t) = \Delta^{\text{rev}}(t' \mid t) + \Delta^{\text{irrev}}(t' \mid t),$$

where Δ^{rev} is the contraction recoverable by reinstating Ξ (capability-availability changes) and Δ^{irrev} is the irreversible component (information disclosure, trust degradation, non-reversible poset structural changes). Reinstatement at $t' + 1$ restores Δ^{rev} to zero by construction (the restriction re-blocks availability-based pathways), leaving $\text{vol}_{\text{P}}(G_{t'+1}) = \text{vol}_{\text{P}}(G_t) - \epsilon - \Delta^{\text{irrev}}(t' \mid t)$. By Definition 4(iii), $\Delta^{\text{irrev}}(t' \mid t) \leq \delta^{\max}_{\text{irrev}} \leq \epsilon$, giving $\text{vol}_{\text{P}}(G_{t'+1}) \geq \text{vol}_{\text{P}}(G_t) - 2\epsilon$.

Part (c): scope isolation. Decompose P into $S \sqcup \partial S \sqcup (P \setminus (S \cup \partial S))$. For any $d \in P \setminus (S \cup \partial S)$, d is not in any cooperative capability spanning S and $P \setminus S$, and no pathway in $\Pi(S)$ has prerequisites or intermediate transitions touching d. The test modifies only capability-availability within S (the only capabilities whose status changes), so the induced subgraph of G_t restricted to d's pathway support is unchanged. Since a pathway's exercise probability [Lasser, 2026c, Definition 2] takes G_t as an argument, and the relevant portion of G_t is invariant, the exercise probability of d's pathways is unchanged during the test. For $c \in \partial S$, the forward-reachability closure condition Definition 4(ii) guarantees that the expected contraction from any $\text{Pathway}_{\text{ext}}$ in $P \setminus S$ whose prerequisites include a state change in c is already included in $\text{max_contraction}(S)$. Hence any cross-boundary cascade is bounded by the $\text{max_contraction}(S)$-budget and therefore by ϵ.

A.4 Proof of Lemma 1

Pathway length in $\Pi(S)$ is bounded by L ([Lasser, 2026c, Definition 2]). The number of distinct pathways is at most $|S|^L$: at each of the L steps we choose one of at most $|S|$ capabilities. For each pathway, the contraction magnitude is computable from the volume axioms M1–M6 [Lasser, 2026e, Proposition 1] in $O(|S|)$ time (enumerating the affected capability tuple). Under the simplifying assumption that cooperative-surface forward-reachability depth is bounded by the same constant L that bounds pathway length (satisfied whenever cross-boundary cascades cannot chain more than pathway-length-many cooperative hops before terminating), enumerating the cooperative surface ∂S and its forward-reachability chains is $O(|P|^{L'})$. Taking the maximum across enumerated pathways and adding cooperative-surface terms yields $\text{max_contraction}(S)$ in $O(|S|^L \cdot |S| + |P|^{L'}) = O(|S|^{L+1} + |P|^{L'})$ time, matching Lemma 1's statement. When $L' \leq L$, the second term is dominated and the bound simplifies to $O(|P|^{L+1})$.

A.5 Proof of Proposition 3

Part (a). Let $p_\tau = \Pr(\text{miss in single window of length } \tau \mid A_n = 1)$. Under Assumptions 1–3 with persistent activation, the window decomposes into τ/τ_{mix} independent opportunities (exact if $\tau_{\text{mix}} = 1$, upper-bound if $\tau_{\text{mix}} > 1$). Under Assumption 3(a) (memoryless detectors, $\tau_{\text{mix}} = 1$) the per-step indicators are independent and the formula is exact, not an upper bound. The upper-bound direction in the $\tau_{\text{mix}} > 1$ case (Assumption 3(b), mixing) requires that the temporal-mixing process satisfy a monotone-coupling property: detector misses at temporally adjacent steps are positively correlated, so partitioning into τ/τ_{mix} independent blocks under-estimates rather than over-estimates the joint miss probability. This positive-association property is standard for monotone functions of an underlying mixing process (Esary–Proschan–Walkup-style positive association of threshold detectors, holding whenever the detector indicator is monotone in its input signal—the standard case for threshold-based detection). Each opportunity misses with probability $1 - \delta_{\text{agg}}$. Hence $p_\tau \leq (1 - \delta_{\text{agg}})^{\tau/\tau_{\text{mix}}}$, decreasing in τ.

Part (b). Inverting the bound: if $\beta(\tau_{\text{test}}) \leq \epsilon_\beta$, then $(1-\delta_{\text{agg}})^{\tau_{\text{test}}/\tau_{\text{mix}}} \leq \epsilon_\beta$, i.e., $\tau_{\text{test}} \geq \tau_{\text{mix}} \log \epsilon_\beta / \log(1-\delta_{\text{agg}})$; taking ceilings gives the claim.

Part (c). Cross-substrate channels satisfy Assumption 2 unconditionally by [Lasser, 2026a, Proposition 5], so the product formula is exact. Temporal independence follows from Assumption 3(a) or (b). Restricting the product to cross-substrate channels yields (10).

A.6 Proof of Theorem 2 (full derivation)

We supply the rigorous a.s.-convergence argument. The sketch in Section 6 reduces to the strong law of large numbers; here we verify the conditions and spell out the quantitative rate.

The log-likelihood random walk. Define

$$\ell_n = \log \frac{\Pr(O_n \mid H_1)}{\Pr(O_n \mid H_0)} \in \{\log L_1, \log L_0\},$$

so that $\operatorname{logit} p_R(t_n) = \operatorname{logit} p_R(0) + \sum_{i=1}^n \ell_i$. Under Assumption 6, the O_n are conditionally i.i.d. given the true hypothesis. Under Assumption 8, every test exercises the capability, so the fixed-binary likelihoods of Section 3 (not the p_{ex}-discounted partial-exercise forms of Assumption 8) are the correct ones. Under Assumption 5, those likelihoods match the true data-generating distribution. Under Assumption 4, $\log L_1 \in (0, +\infty)$ and $\log L_0 \in (-\infty, 0)$; both are finite (the likelihoods are bounded strictly away from 0 and 1 because $\alpha_{\text{fp}} \in (0,1)$ and $\beta \in (0,1)$ imply $\Pr(O=0 \mid H_1), \Pr(O=1 \mid H_0) > 0$ and strictly less than 1; the non-degenerate-detector regime of Assumption 4 rules out the boundary cases $\alpha_{\text{fp}} = 0$ and $\beta = 0$). Hence ℓ_n is a bounded i.i.d. sequence.

Drift under each hypothesis. Under the true hypothesis H_j,

$$\mathbb{E}_{H_j}[\ell_n] = \Pr(O=1 \mid H_j) \log L_1 + \Pr(O=0 \mid H_j) \log L_0.$$

Under H_1 this equals $\mathrm{KL}(\Pr(\cdot \mid H_1) \,\|\, \Pr(\cdot \mid H_0)) > 0$; under H_0 it equals $-\mathrm{KL}(\Pr(\cdot \mid H_0) \,\|\, \Pr(\cdot \mid H_1)) < 0$. Both are strictly nonzero by Assumption 4 (the two hypothesis-conditional distributions differ, so their KL divergence is strictly positive).

Strong law of large numbers. By the strong law (e.g., [Cover and Thomas, 2006, Ch. 6]) applied to the bounded i.i.d. sequence $\{\ell_n\}$,

$$\frac{1}{n} \sum_{i=1}^n \ell_i \xrightarrow{\text{a.s.}} \mathbb{E}_{H_j}[\ell_n]$$

under the true hypothesis H_j. Hence $\sum_{i=1}^n \ell_i \to +\infty$ a.s. under H_1 and $\to -\infty$ a.s. under H_0. Translating to p_R via the logistic map ($p_R = \sigma(\operatorname{logit} p_R)$), which is well-defined at the initial point because $p_R(0) \in (0,1)$ by Assumption 7 (so $\operatorname{logit} p_R(0)$ is finite), $p_R(t_n) \to 1$ a.s. under H_1 and $\to 0$ a.s. under H_0. This establishes Part (a).

Part (b) (run under H_0). Conditional on all $O_i = 0$, each $\ell_i = \log L_0$, so $\sum_{i=1}^n \ell_i = n \log L_0$. Hence $\operatorname{logit} p_R(t_n) = \operatorname{logit} p_R(0) + n \log L_0$, and $p_R(t_n) \to 0$ at geometric rate $|\log L_0|$. The unconditional statement under H_0 simply replaces "conditional on all $O_i = 0$" with an expected drift $(1-\alpha_{\text{fp}}) \log L_0 + \alpha_{\text{fp}} \log L_1$, which is $-\mathrm{KL}(\Pr(\cdot \mid H_0) \,\|\, \Pr(\cdot \mid H_1))$; again strictly negative.

Part (c) (rate under H_1). Each test produces $O_n = 1$ with probability $q_1 = \Pr(O=1 \mid H_1) = \theta_1(1-\beta) + (1-\theta_1)\alpha_{\text{fp}}$. The first detection is a geometrically distributed stopping time τ_1 with mean $1/q_1$. The log-odds update on $O=1$ is $\log L_1 > 0$. For θ_1 close to 1 and small β, a single detection update $\operatorname{logit} p_R + \log L_1$ exceeds any fixed threshold $\operatorname{logit}(1-\delta)$, so the expected time to reach $p_R > 1-\delta$ is $O(1/q_1) = O(1/[\theta_1(1-\beta)])$ as stated.

Part (d) (KL form). The drift-rate statements above rewrite to

$$\begin{aligned} H_0: \ -\mathbb{E}_{H_0}[\ell_n] &= (1-\alpha_{\text{fp}})\log(1/L_0) + \alpha_{\text{fp}}\log(1/L_1) \\ &= \mathrm{KL}(\Pr(\cdot \mid H_0) \,\|\, \Pr(\cdot \mid H_1)), \\ H_1: \ \mathbb{E}_{H_1}[\ell_n] &= \Pr(O=1 \mid H_1)\log L_1 + \Pr(O=0 \mid H_1)\log L_0 \\ &= \mathrm{KL}(\Pr(\cdot \mid H_1) \,\|\, \Pr(\cdot \mid H_0)). \end{aligned}$$

Both are strictly positive under Assumption 4, so Chernoff's large-deviations result [Chernoff, 1952] gives the corresponding exponent on the probability of misclassification: $\Pr(p_R(t_n) < 1/2 \mid H_1)$ decays at least as fast as $\exp(-n \cdot \mathrm{KL}_{H_1})$ for some $\mathrm{KL}_{H_1} > 0$, and symmetrically for H_0.

A.7 Proof of Corollary 3

Under H_0 with all n tests producing $O_n = 0$, $\operatorname{logit} p_R(t_n) = \operatorname{logit} p_R(0) + n \log L_0$. Setting $\operatorname{logit} p_R(t_n) \leq \operatorname{logit} p_{\text{clear}}$ and solving for n:

$$n \geq \frac{\operatorname{logit} p_{\text{clear}} - \operatorname{logit} p_R(0)}{\log L_0}.$$

Using $\operatorname{logit} x = \log(x/(1-x))$, the numerator equals $\log(p_{\text{clear}}(1-p_R(0))/(p_R(0)(1-p_{\text{clear}})))$. Note $\log L_0 < 0$, so the inequality direction flips when dividing; taking the ceiling gives (13).

A.8 Proof of Proposition 7

Part (a). Axiom M4 (weak monotonicity) of $\mathrm{vol}_{\mathrm{P}}$ [Lasser, 2026e, Proposition 1] gives that adding capabilities to P (equivalently, lifting a restriction) never decreases $\mathrm{vol}_{\mathrm{P}}$ ignoring risk effects. Subtracting the worst-case risk effect, bounded by ϵ during the test and by 2ϵ post-cutoff (Theorem 1), gives the stated interval.

Part (b). The restriction criterion [Lasser, 2026c, Proposition 1] depends on p_R; replacing the pre-test p_R with the post-test $p_R(t + \tau_{\text{test}})$ produces a criterion evaluated against a more accurate estimate. If $p_R(t+\tau_{\text{test}}) < p_{\text{thresh}}$, the criterion no longer fires and the restriction is lifted; otherwise it is maintained.

Part (c). Theorem 2 establishes that $p_R(t_n) \to \mathbf{1}_{H_1}$ a.s. for each tuple. Since the restriction criterion is continuous in p_R and restricts iff $p_R \geq p_{\text{thresh}}$, the restriction decisions converge to $\{\text{restrict} : \text{genuinely risky}\}$ a.s.

B Notation Summary

Table 3 collects the principal symbols specific to this paper. Symbols inherited from earlier papers [Lasser, 2026e,c,f,a,d]—including $\mathrm{vol}_{\mathrm{P}}$ (capability volume), G_t (global capability graph), P, P_t (capability poset), $\mathrm{Pathway}$ (exercise pathway), $\mathcal{R}$, $\mathcal{R}_j$ (concentration risk), a_j (agent index), and T_j^{risk} (risk-trust factor)—retain their original meanings and are not redefined here.

Symbol	Definition	Introduced in
Risk-restriction object		
R	Risk claim $(a_j, (d_1, \ldots, d_k), \mathrm{Pathway}, p_R)$	Def. 1
Ξ	Preemptive restriction of a keystone capability	Def. 1
p_R, $p_R(t)$	Posterior probability the risk claim is valid	Def. 1
p_{thresh}	Restriction-firing threshold for p_R	Prop. 2
Controlled relaxation test		
T	CR test tuple $(R, \Xi, S, \tau_{\mathrm{test}}, \epsilon)$	Def. 3
$S \subseteq P$	Test scope (bounded sub-poset)	Def. 3, 4
$\tau_{\mathrm{test}} \in \mathbb{N}$, τ_{opt}, τ_{min}	Test duration; info-value-optimal; target-ϵ_β minimum	Def. 3, Prop. 6, 3
ϵ, ϵ_{user}, $\epsilon_{\mathrm{total}}$	Damage tolerance; user ceiling; aggregate budget	Def. 3, 6, 7
$\mathrm{max_contraction}(S)$, ∂S	Max. $\mathrm{vol}_{\mathrm{P}}$-contraction in S; cooperative surface	Lem. 1
δ_{irrev}, $\delta_{\mathrm{irrev}}^{\mathrm{max}}$	Irreversible pathway fraction; maximum within scope	Def. 4(iii)
Binary observation model and detection		
A_n, $O_n \in \{0, 1\}$	Latent activation; observed contraction at test n	Sec. 3
H_0, H_1	Pathway-harmless ($\theta = 0$) vs. -dangerous ($\theta = \theta_1$)	Sec. 3
θ_1, β, α_{fp}	Activation rate (H_1); false-negative; false-positive	Sec. 3
L_0, L_1	Per-test likelihood ratios for $O = 0$, $O = 1$	Eq. (3)
δ_i, δ_{agg}	Per-channel; aggregate detection probability	Def. 5, Eq. (6)
δ_{cross}, $\delta_{\mathrm{agg}}^{\mathrm{cons}}$	Cross-substrate; conservative aggregate	Prop. 3(c), Eq. (7)
τ_{mix}, k_{act}	Detector mixing time; activation duration	Ass. 1–3
Convergence, meta-policy, refinement		
KL	Kullback–Leibler divergence	Thm. 2(d)
N_{clear}, p_{clear}	Tests required to clear a false claim; target posterior	Cor. 3
$\mathrm{IV}(T)$, T_{stale}	Expected information value; minimum re-test age	Def. 6
Σ	Test schedule	Def. 7
K	Keystone structure realising Ξ	Def. 8
$K_1 \preceq K_2$, $K \vdash \mathrm{Pathway}$	Keystone partial order; "blocks pathway" relation	Def. 8–9

Table 3: Principal notation specific to this paper.

C Cross-Paper Dependencies

Table 4 summarizes which results from [Lasser, 2026b,e,c,f,a,d] this paper relies on and how they enter the analysis.

Source	Result used	Role here
Poset	Axiom verification (Prop. 1), benchmark refinement	$\mathrm{vol}_{\mathrm{P}}$ properties preserved under relaxation; complementary evidence source
Horizon	Exercise Pathway (Def. 2), Concentration Risk (Def. 3), Risk Claim (Def. 4), Preemptive Restriction (Prop. 1)	The risk machinery the protocol operates within
Horizon	Structural discovery value (Prop. 2), anti-monopolar property (Prop. 6)	Value framework for test-cost vs. restriction-cost comparison
Causal	Risk-Trust Dynamics (Def. 4), Log-Odds Risk Aggregation (Def. 5), L^2 Convergence (Prop. 2)	Bayesian update machinery that processes test outcomes
Causal	Multi-channel architecture (Table 1), Causal Attribution (Def. 2)	Formalizing what "observing the outcome" means during a test
Verification	Verification Asymmetry (Def. 2), Monotonicity (Prop. 1), Exercise Protocol (Def. 12), Risk Claim Protocol (Def. 13)	The formal verification-asymmetry framework the protocol resolves
Verification	Channel Isolation (Prop. 5), Channel Independence (Prop. 6)	Integrity conditions on test observations
Phase	Phase boundary (Thm. 1), Wamura problem as degeneracy (Prop. 4)	Controlled relaxation makes $r_{\mathrm{S}} > 0$ for Loop 3

Table 4: Dependencies on prior results. Poset = Lasser [2026e], Horizon = Lasser [2026c], Causal = Lasser [2026f], Verification = Lasser [2026a], Phase = Lasser [2026d].